
Design and Development of Multilevel Converter Fed Induction Motor Drive For Medium Power Application

A Report submitted for Undergraduate Project (UGP)
in Bachelor of Technology

by

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to the

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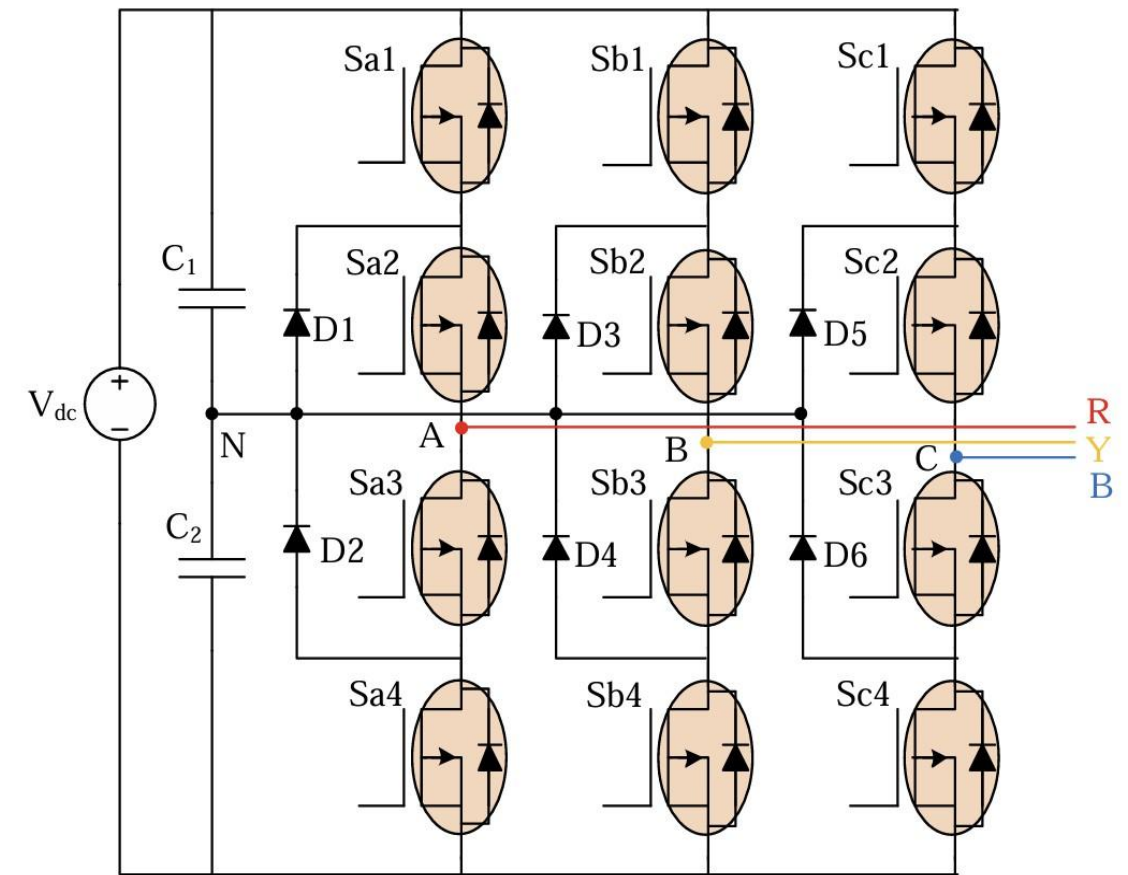
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Designing And Implementation of SiC MOSFET Based NPC Converter

This work presents the design of a **compact, robust, and efficient three-level three-phase Neutral Point Clamped (NPC) converter** using **SiC MOSFET technology** for medium-power industrial and traction motor drive applications. It focuses on the design and implementation of a **700 V DC, 6.6 kW NPC converter**, where the high switching frequency of **100 kHz** helps reduce filter size and electromagnetic interference (EMI) while improving overall system efficiency.

Key Parameters

- Input DC Link Voltage: 700V
- Rated Power: 6.6kW
- Switching Frequency: 100kHz
- Output AC Voltage: 415V
- DC-Link Capacitor: 680 μ F



Schematic of 3-Level NPC inverter

Background and Motivation

The Challenge with Conventional Inverters

- Conventional **2-level voltage source inverters** have high dv/dt , larger harmonic distortion, and higher instantaneous errors, which create stress on switching devices.
- These inverters are limited in **voltage level, power level, and switching frequency**, and they experience higher conduction and switching losses, reducing converter efficiency.
- For **smooth and efficient motor drive operation**, the inverter output should have fewer harmonics and be closer to a sinusoidal waveform to ensure better power regulation.

The Multilevel Solution

- **Multilevel inverters** produce output with lower harmonics and lower dv/dt , which reduces stress on switching devices.
- Compared to **2-level inverters**, they offer lower switching and conduction losses, support higher power levels, and generate a waveform closer to a sinusoidal output.
- Although multilevel inverters increase circuit complexity, they overcome the limitations of 2-level inverters; the most common topology used is **NPC (Neutral Point Clamped)**.
- Some techniques are
 1. PWM Control Techniques
 2. Advanced Methods
 3. CMVI Benefits

1. PWM Control Techniques

PWM techniques control switching devices to generate variable voltage and frequency; **SPWM** is the most commonly used method.

In medium and high-power applications, **balancing the DC-link capacitor** in multilevel inverters becomes difficult.

2. Advanced Methods

Other modulation methods include **Square Wave PWM, SVPWM, third harmonic injection, CMVI**.

Third harmonic injection or **Common Mode Voltage Injection** helps solve the **DC-link capacitor voltage balancing** problem.

3. CMVI Benefits

CMVI helps reduce unwanted currents caused by common mode voltage in inverters.

This reduces **motor damage and bearing erosion**, improving reliability.

In a **3-level NPC inverter**, the pole voltage has three states:

Positive State (Sa1 & Sa2 ON $\rightarrow V_{AO} = +V_{dc}$

Zero State (D1, D2 with Sa2 & Sa3 ON $\rightarrow V_{AO} = 0$

Negative State (Sa3 & Sa4 ON $\rightarrow V_{AO} = -V_{dc}$

The switches generate the **positive and negative voltage levels**, while **clamping diodes create the zero voltage level**.

Switch pairs **Sa1–Sa3** and **Sa2–Sa4** operate **complementarily**, and this arrangement is used in all three inverter legs.

Switching State	Sa1	Sa2	Sa3	Sa4	Output Pole Voltage
Positive (+)	1	1	0	0	+Vdc
Zero (0)	0	1	1	0	0
Negative(-)	0	0	1	1	-Vdc

Multilevel Inverter Architecture

Classification and NPC Topology

- **Neutral Point Clamped (NPC) MLI:**

Uses clamping diodes to produce multiple voltage levels and provides good voltage balancing; widely used in traction drives and EV charging

- **Flying Capacitor (FC) MLI:**

Generates voltage levels using capacitors but requires capacitor voltage balancing, which makes the circuit design more complex.

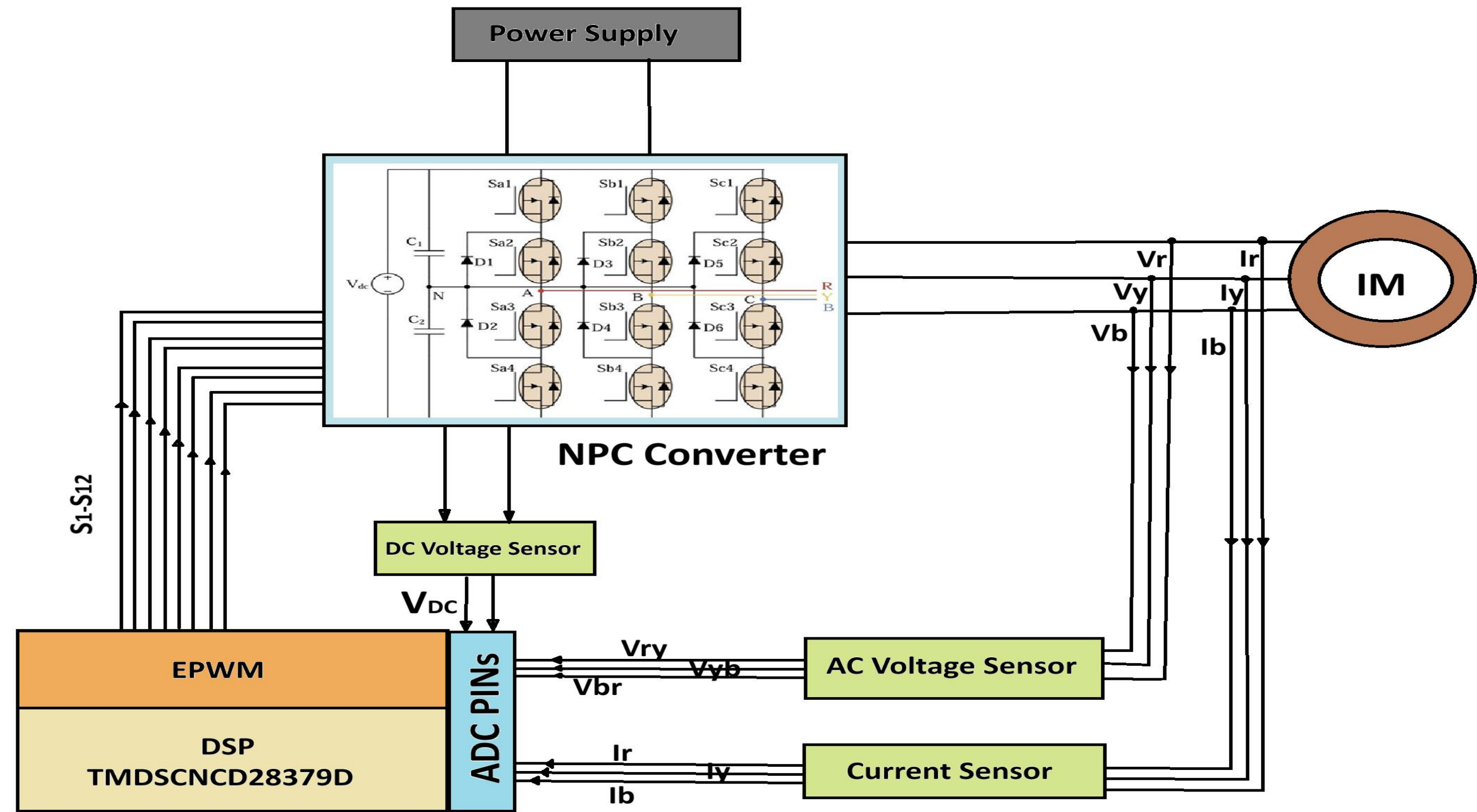
- **Cascaded H-Bridge (CHB) MLI:**

Uses separate isolated DC sources for each H-bridge module; offers modular structure but requires more components

3-Level NPC Working Principle

- A **3-phase 3-level NPC inverter** uses four active switches (S_{a1} – S_{a4}) and four antiparallel diodes in each phase leg; the switches can be **IGBTs or MOSFETs**.
- **Six clamping diodes** are used per phase leg, and each switch blocks **half of the input DC voltage**.
- **Capacitors C1 and C2** form the DC-link to balance voltage, with **N as the neutral point** and the voltage between **A and N called pole voltage**.
- **Diodes D1 and D2** are connected to the neutral point, and the **DC-link voltage must remain balanced** to avoid stress on switching devices.
- The inverter has **$3^3=27$ possible switching combinations**, producing different output voltage.

Schematic Digram of 3 Phase 3 Level NPC Converter fed IM Drive in Closed Loop



Steps for Converter Design

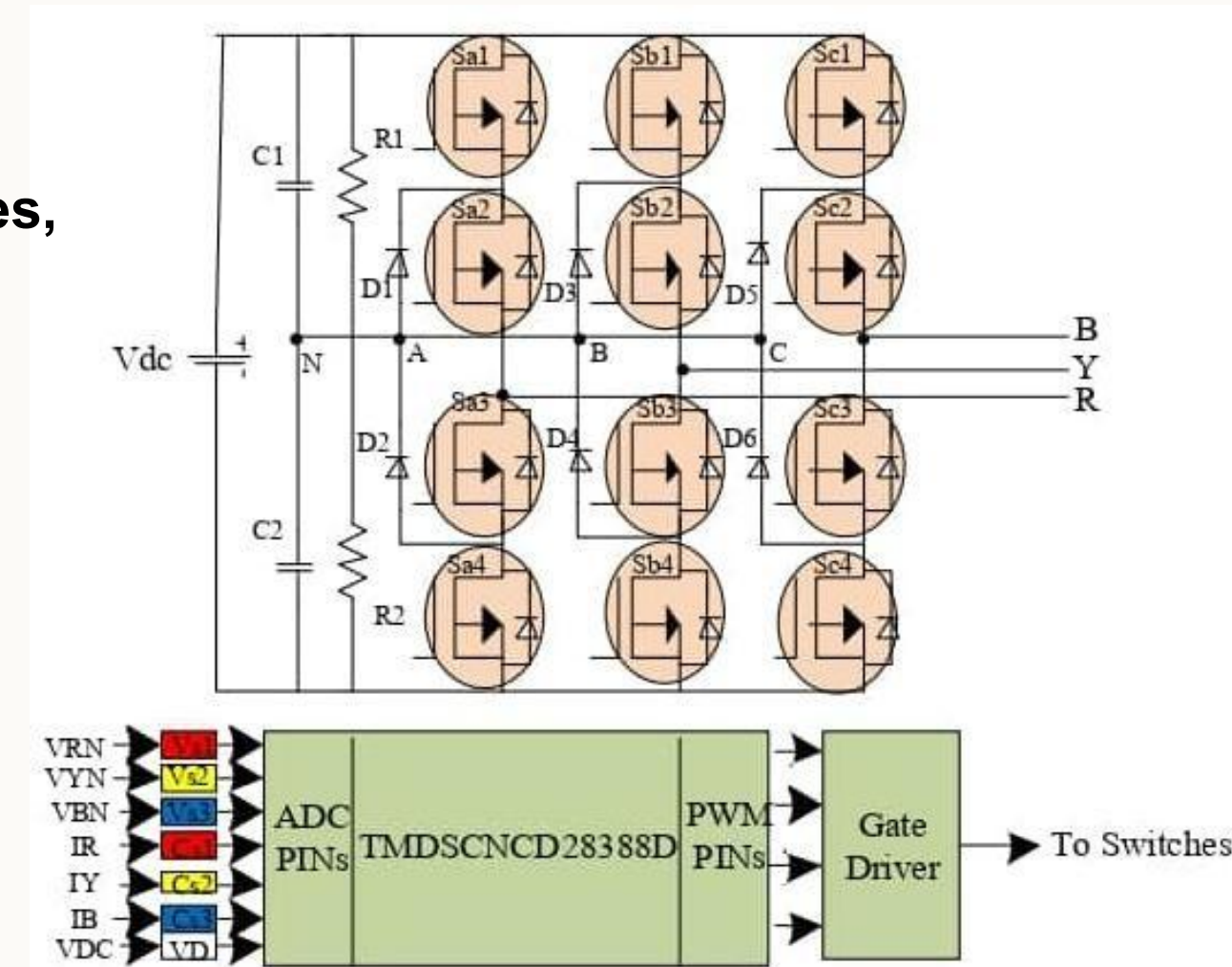
The circuit diagram and specification of three-phase SiC MOSFET based NPC converter is shown below

Design Approach

The design process includes careful selection of **switching devices**, **gate drivers**, and **sensors**. All circuits were initially **simulated in LTspice** and then implemented on **separate PCBs** to ensure a **compact, reliable, and efficient converter operation**.

Specification	Value
Rated Power (POUT)	6.6 kW
DC-Link Voltage (V_{dc})	700 V
DC-Link Capacitance (C_{dc})	680uF
Bleeder Resistance (R_b)	100 k Ω
Line-Line Output Voltage (V_{RY})	415 V
Switching Frequency (F_{sw})	100 kHz

Converter Specification



3-Level NPC Converter

Steps for Converter Design

■ Selection of Components-

Selection of SiC MOSFET Switch-

- The first step in converter design is selection of switch based on required voltage and current rating.
- The output equation is-

$$P_{out} = \sqrt{3}V_L I_L \cos\phi$$

Here for 6.6kW power rating, line voltage taken 415V and a nominal load power factor 0.85.
so $I_L = 10.81A$ and peak current=15A

- Based on voltage and current rating we calculated loss for each switch.
- Select switch with lower losses

Loss Calculation of MOSFET-

- Conduction loss is calculated as-

$$P_{CM} = R_{DS(on)} I_D^2 \left(\frac{1}{8} + \frac{m_a \cos\phi}{3\pi} \right) = 4.38W$$

- Switching loss is calculated as-

$$P_{SWM} = \frac{1}{\pi} (E_{ON} + E_{OFF}) \frac{V_{DC}}{V_{DC,TEST}} \frac{I_o}{T_{TEST}} F_{SW} = 5.95W$$

Choosing the **appropriate power switch** is essential for achieving good converter performance. Six **SiC MOSFETs from different manufacturers** were analyzed based on their **voltage rating (>350 V to block half of the DC voltage)**, **current capability (>15 A peak)**, and **total losses including body diode losses**.



Loss Calculation of body diode-

- Conduction loss is calculated as-

$$P_{CM} = U_{DO} I_o \left(\frac{1}{2\pi} - \frac{m_a \cos\phi}{8} \right) + R_D I_o \left(\frac{1}{8} - \frac{m_a \cos\phi}{3\pi} \right) = 1.41W$$

- Switching loss is calculated as-

$$P_{SWM} = \frac{1}{\pi} (E_{ON} + E_{OFF}) \frac{V_{DC}}{V_{DC,TEST}} \frac{I_o}{T_{TEST}} F_{SW} = 3.31W$$

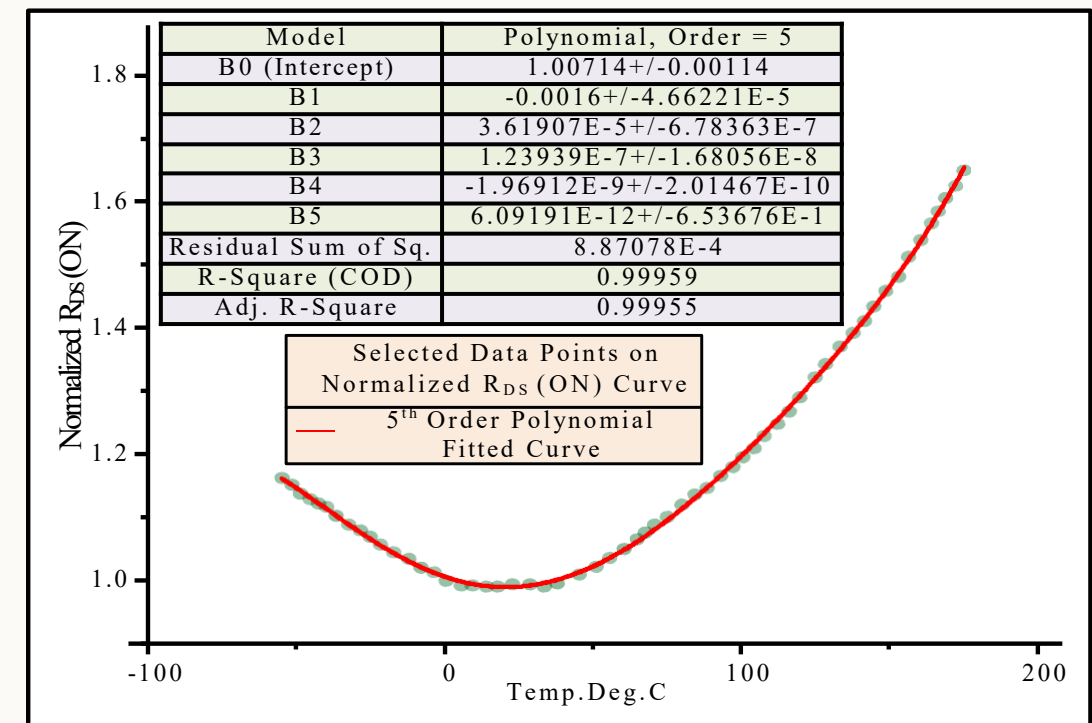
Steps for Converter Design

- The $R_{DS(ON)}$ calculation for converter design is shown in Fig. 2.
- The switches comparison on loss calculation and cost is given in Table.
- The selected switch for designed converter is **TW083N65C**.

This switch is selected because it has the minimum loss

S.N o	Switch	Total Loss
1.	TW083N65C	15.5 W
2.	IMZA120R040M1H	16.8 W
3.	NVH4L030N120M3S	18.54 W
4.	SCT4036KRHR	17.23 W
5.	G3R40MT12K	18.5 W
6.	IMZ120R030M1H	21.58 W

Switch Comparison



$R_{DS(ON)}$ Calculation from Origin Software

Steps for Converter Design

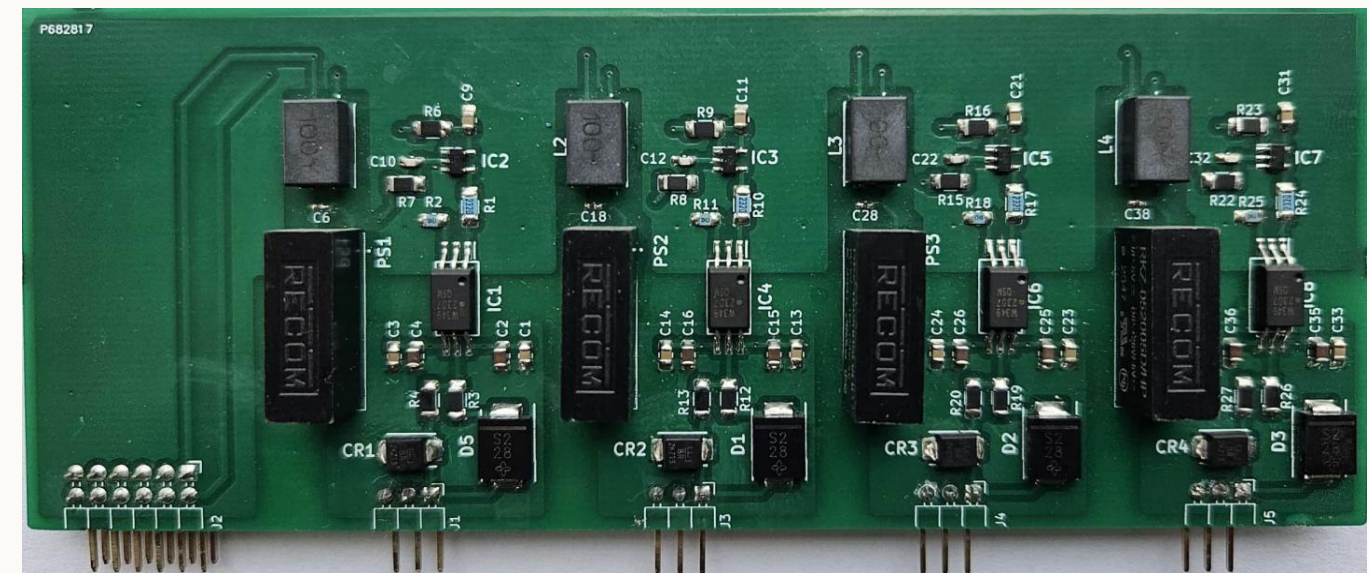
Selection of gate driver-

- While selecting gate driver certain point should be taken care.
 - a) Isolation between input and output.
 - b) Under voltage lockout (ULVO) protection.
 - c) Active miller clamping.
 - d) Proper selection of gate resistance.

Gate resistance calculation -

- V_{CC} and V_{EE} are obtained from switch data sheet. $I_{o(Peak)}$ is maximum gate current from gate driver data sheet.
- Selection of gate driver based on switch's turn on and off voltage.
- For the NPC converter design, we have selected **ACPL-349-00E** gate driver.

$$R_G \geq \frac{V_{CC} - V_{EE}}{I_{o(Peak)}} - R_{Gint} = \frac{20 - (-5)}{2.5} - 0.5 \geq 9.5\Omega$$



Gate Driver PCB

Gate Resistance Calculation and Component Selection

External gate resistance R_g and internal minimum turn-on resistance R_{gint} ensure output current stays within device absolute maximum ratings. The calculation follows the gate driver IC datasheet specifications.

Calculation Formula

$$R_g \geq \frac{V_{CC} - V_{EE}}{I_{Opeak}} - R_{gint}$$

Where $V_{CC} = +20V$, $V_{EE} = -5V$,
 $I_{Opeak} = 2.5A$, $R_{gint} = 0.5\Omega$

Calculated Value

$$R_g \geq \frac{20 - (-5)}{2.5} - 0.5 = 9.5\Omega$$

Selected value: **10 Ω** for safe
operation with minimized power loss

Selected Components

Gate Driver IC: ACPL-349-000E

Peak Current: 2.5A (4A max)

Output Range: +20V/-5V

The Broadcom ACPL-W349 high-speed optocoupler delivers peak output current up to 2.5A, making it ideal for driving wide bandgap semiconductors like SiC and GaN MOSFETs, as well as IGBTs. Simulation results validate the design, showing proper +20V/-5V output from 0-3.3V MCU input at 100kHz switching frequency.

Steps for Converter Design

Selection of DC-Link Capacitor-

- The DC-Link capacitor value is calculated by taking 5% ripple in DC-Link voltage and it is given as-

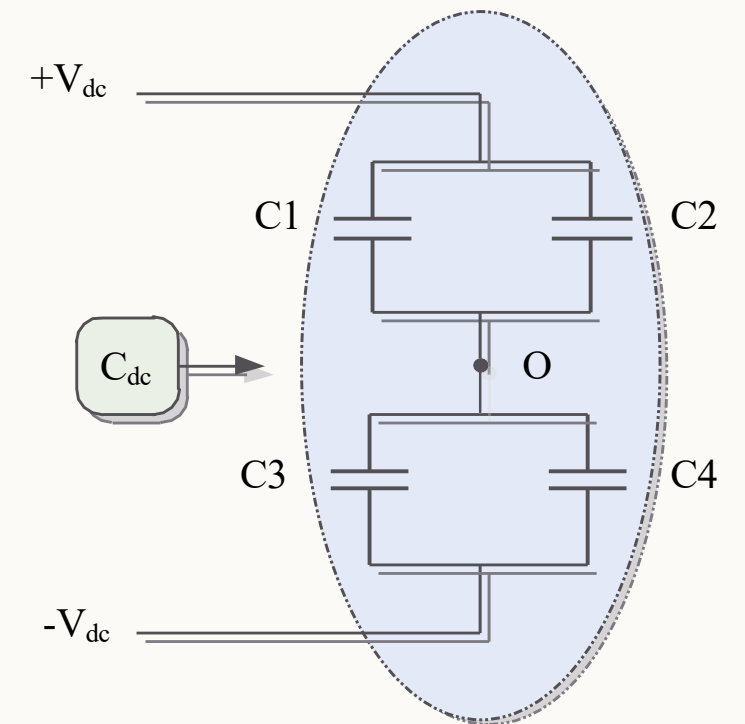
$$C_{dc} = \frac{2P}{F_{SW}(\Delta V)^2} = 586\mu F$$

- If the capacitor value is not available, then take a series/parallel combination or both can be taken
- The combination should have minimum ESR and ESL.



DC-Link Capacitor

Four 680 μ F capacitors in series-parallel configuration provide voltage stability and ripple reduction. Calculated minimum value is 583.3 μ F for 5% voltage ripple at 100kHz switching frequency with 6.6kW power.



DC-Link Capacitor

Selection of heat sink -

- Thermal resistance model for heat sink is given as-

$$R_{th} = \frac{T_j - T_a}{P_D} - R_{th,JC} - R_{th,CS} = 3.657^\circ\text{C}/\text{W}$$

Here,

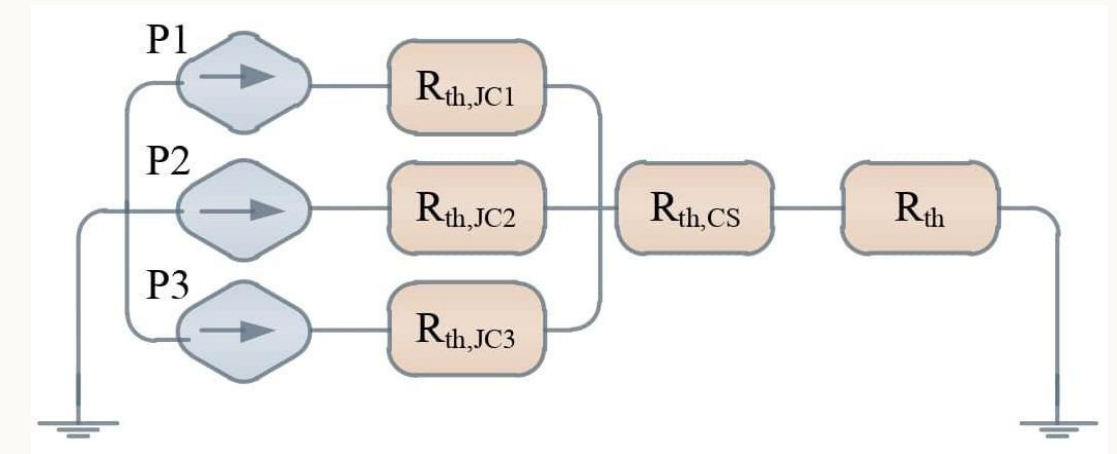
$$R_{th,JC1} = R_{th,JC2} = R_{th,JC3} = 1.35^\circ\text{C}/\text{W}$$

Maximum junction temperature (T_j) = 175°C

Ambient temperature (T_a) = 25°C

$$R_{th,CS} = 0.5^\circ\text{C}/\text{W}$$

$$P_D = P_1 + P_2 + P_3 = 32.55\text{W}$$



Thermal-Resistance Model of Heat Sink

Thermal Management and Heat Sink Selection

- Effective thermal management is essential to dissipate heat from high-power devices and ensure converter reliability and efficiency.
- Total power loss of **32.55 W** with **T_j = 175 °C** and **T_a = 25 °C** requires a heat-sink thermal resistance of **≤ 3.65 °C/W**.
- CR201-75AE heat sink (2.6 °C/W)** is selected, providing sufficient thermal margin for safe operation.



Steps for Converter Design

Auxiliary Power Supply Design-

- To provide power supply to all sensor and gate driver ICs, it is necessary to design power supply.
- In this power supply design, a 15W, 230V AC to 5V DC Converter is used to supply 5V w.r.t. microcontroller ground.
- A 5V-3.3V Isolated DC-DC Converter is used to supply 3.3V w.r.t. V_{DC} .

1

Stage 1: AC-DC Conversion

LS13-13B05DR3 converts 230V AC to 5V DC with pi filter for smoothing. TVS diode SMBJ 7.0A provides OV protection.

2

Stage 2: Isolated DC-DC

TPS73733DCQR creates isolated 3.3V supply with separate ground for current sensors and differential amplifiers.

3

Stage 3: Gate Driver Supply

LD15-23B05R2 provides 15W, 5V DC supply that generates +20V and -5V for gate driver circuits.

S. No.	Components	Voltage
1.	Gate Driver	5V
2.	AC Voltage Sensor	3.3 V
3.	DC Voltage Sensor	5 V
4.	Current Sensor	5V

Voltages Requirement of ICs



PCB Layout of Power Supply

Steps for Converter Design

Snubber Circuit Design-

- Here, we are using RCD snubber as given in Fig. 12.
- The calculation of relevant parameters are calculated as-

$$C > \frac{L_m \times I_o^2}{V_{DS_Surge}^2 - V_{DC}^2} > 66pf$$

And,

$$R \leq \frac{1}{2} \sqrt{\frac{L_m}{C}} \leq 16.6\Omega$$

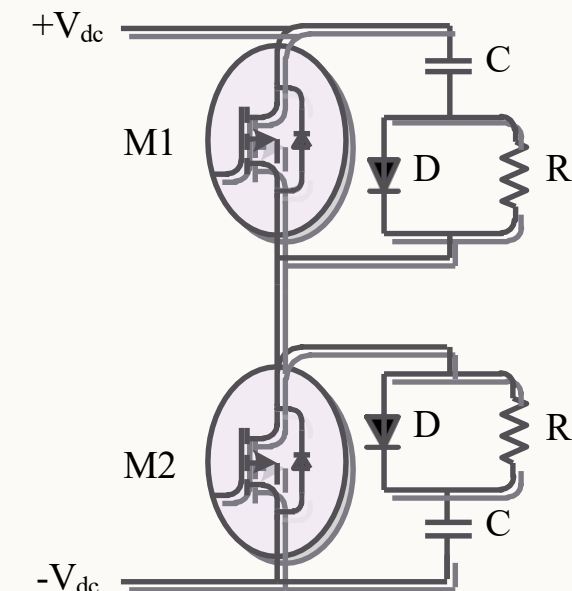
And power rating of resistance is calculated as-

$$P = \frac{F_{SW} \times C \times V_{DS_Surge}^2}{2}$$

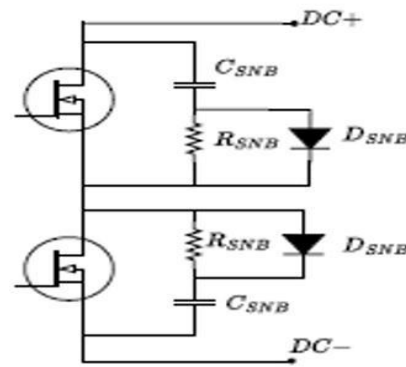
Here, $C = 100pF$, $V_{DS_Surge} = 520V$, $L_m = 110nH$, $V_{DC} = 700V$, $R = 15\Omega$ and diode is Schottky diode of sufficient voltage blocking capacity.

RCD Snubber Circuit

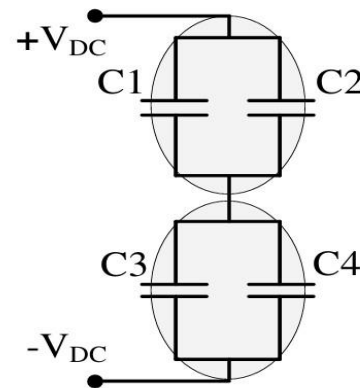
Protects power switches from voltage spikes using $R=15\Omega$ and $C=100pF$ with a fast recovery diode. The snubber capacitance must exceed $66pF$, and resistance should be less than 16.6Ω with power dissipation of $0.967W$.



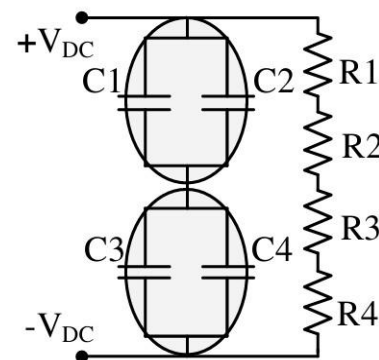
Snubber Circuit



RCD Snubber Circuit



DC Link Capacitor Schematic



Bleeder Resistor Schematic

Protection Circuits and Components



RCD Snubber Circuit

Protects power switches from voltage spikes using $R=15\Omega$ and $C=100\text{pF}$ with a fast recovery diode. The snubber capacitance must exceed 66pF , and resistance should be less than 16.6Ω with power dissipation of 0.967W .



DC-Link Capacitor

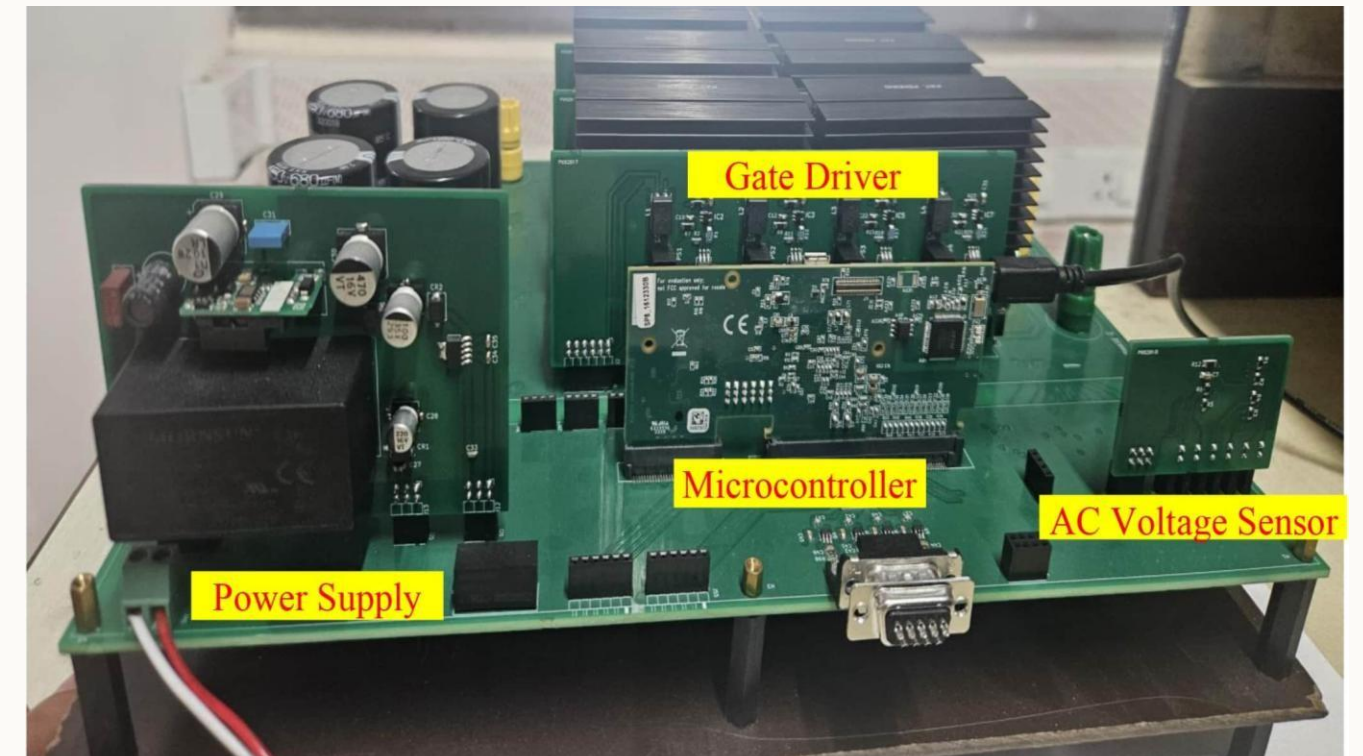
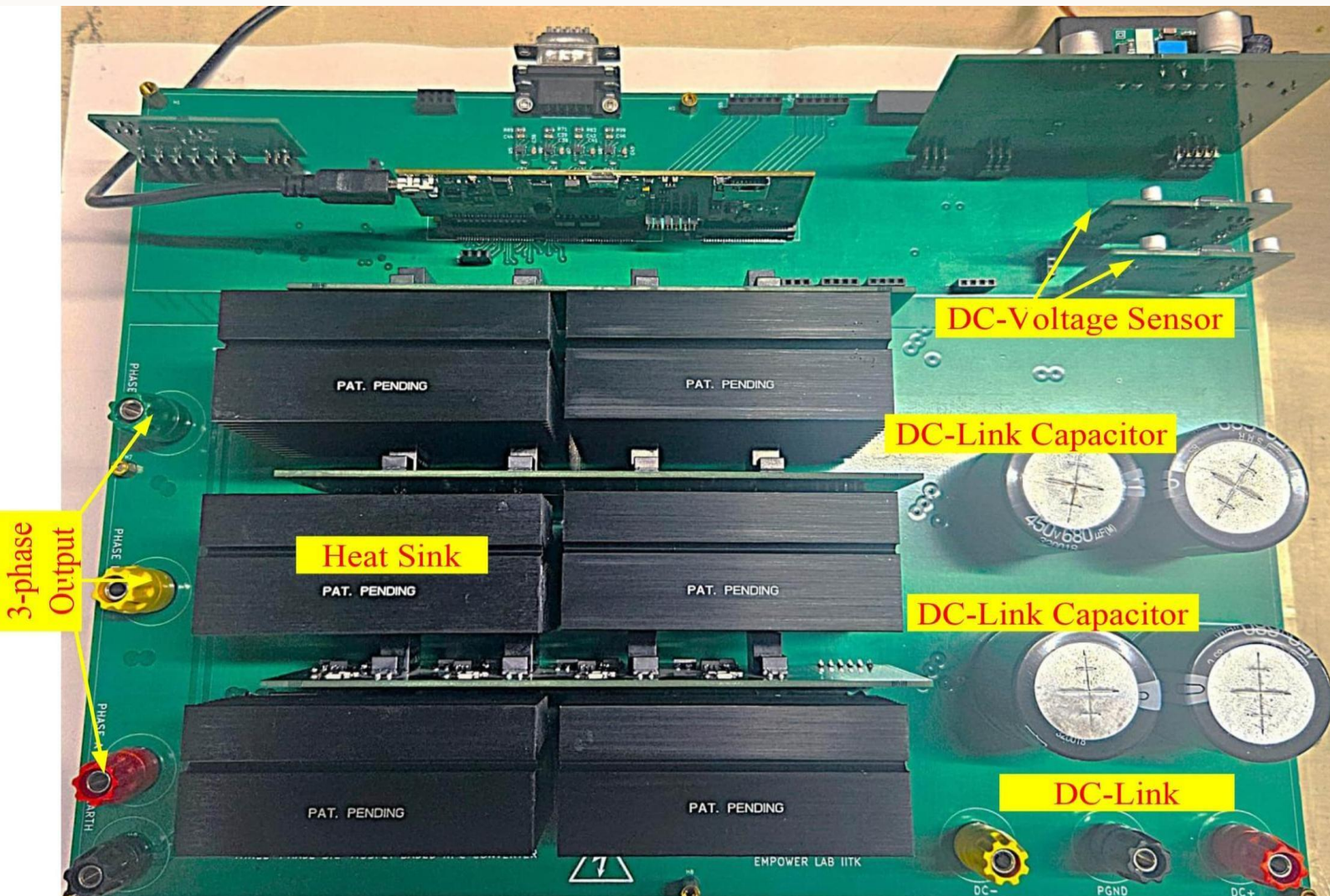
Four $680\mu\text{F}$ capacitors in series-parallel configuration provide voltage stability and ripple reduction. Calculated minimum value is $583.3\mu\text{F}$ for 5% voltage ripple at 100kHz switching frequency with 6.6kW power.



Bleeder Resistor

$100\text{k}\Omega$ resistors discharge DC-link capacitors safely when power is removed, preventing electric shock and component damage. High resistance value minimizes power loss during normal operation.

Hardware Setup of Designed SiC MOSFET Based 3-Phase 3-Level NPC Converter



Designed NPC Converter

Design Summary and Key Achievements

Optimized Switch Selection

TW083N65C from Toshiba selected for minimum losses (15.05W) among six evaluated devices, ensuring efficiency operation at 100kHz switching frequency.

Comprehensive Protection

RCD snubber circuits, DC-link capacitors (680 μ F), and bleeder resistors (100k Ω) provide robust protection against voltage spikes and ensure safe discharge.

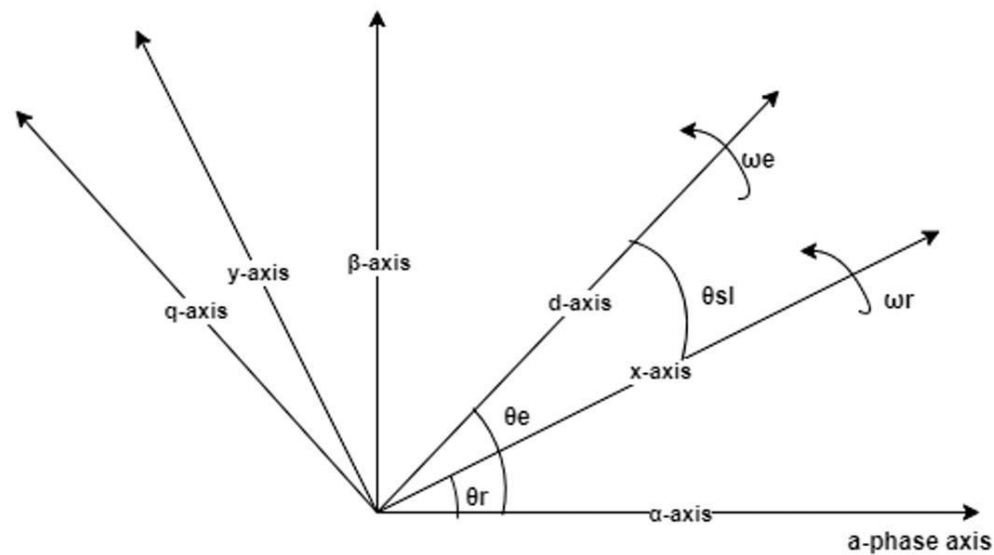
Compact Power Supply

93mm $\times$ 72mm auxiliary power supply delivers multiple isolated voltages (5V, 3.3V, $\pm$ 20V) with high efficiency and minimal footprint.

Effective Thermal Management

CR201-75AE heat sink with 2.6 $^{\circ}$ C/W thermal resistance ensures reliable operation under full load conditions with adequate safety margin.

This chapter presented a comprehensive design methodology for a SiC MOSFET-based three-level NPC converter, covering component selection, loss calculations, protection circuits, power supply design, and thermal management. The systematic approach ensures efficient, compact, and robust converter operation for 6.6kW power conversion applications.



Reference Frames

Vector Control of IM fed from SiC MOSFET based NPC Converter

This chapter presents the hardware implementation and testing of a Neutral Point Clamped (NPC) converter using SiC MOSFETs for induction motor control. The system employs advanced Vector Control techniques to achieve precise speed and torque regulation with superior dynamic response.

Vector Control Architecture

1. Vector Control, also known as **Field Oriented Control (FOC)**, provides excellent dynamic response in motor drives.
2. It works by **independently controlling torque and flux** of the induction motor.
3. The control system uses a **dual-loop structure**:
 - Inner current loop** (fast response)
 - Outer speed loop** (slower response)
4. The operating frequencies of the inner and outer loops differ by **approximately 10 decades**.
5. The **outer speed loop generates the torque reference**, which is used to calculate the **q-axis current** I_{qs} .
6. **Torque control** is achieved through the **q-axis component** of current.
7. **Flux control** is achieved through the **d-axis component** of current.

This decoupling allows **independent control of motor speed and flux**, improving performance and stability.

Steps for Control System Implementation are

1. Signal Acquisition
2. Digital Processing
3. PWM Generation

1. Signal Acquisition

- Encoder measures the **motor speed**.
- Current sensors detect **three-phase motor currents**.
- **ADC pins of TMDSCNCD28379D** convert analog sensor signals into **digital signals**.

2. Digital Processing

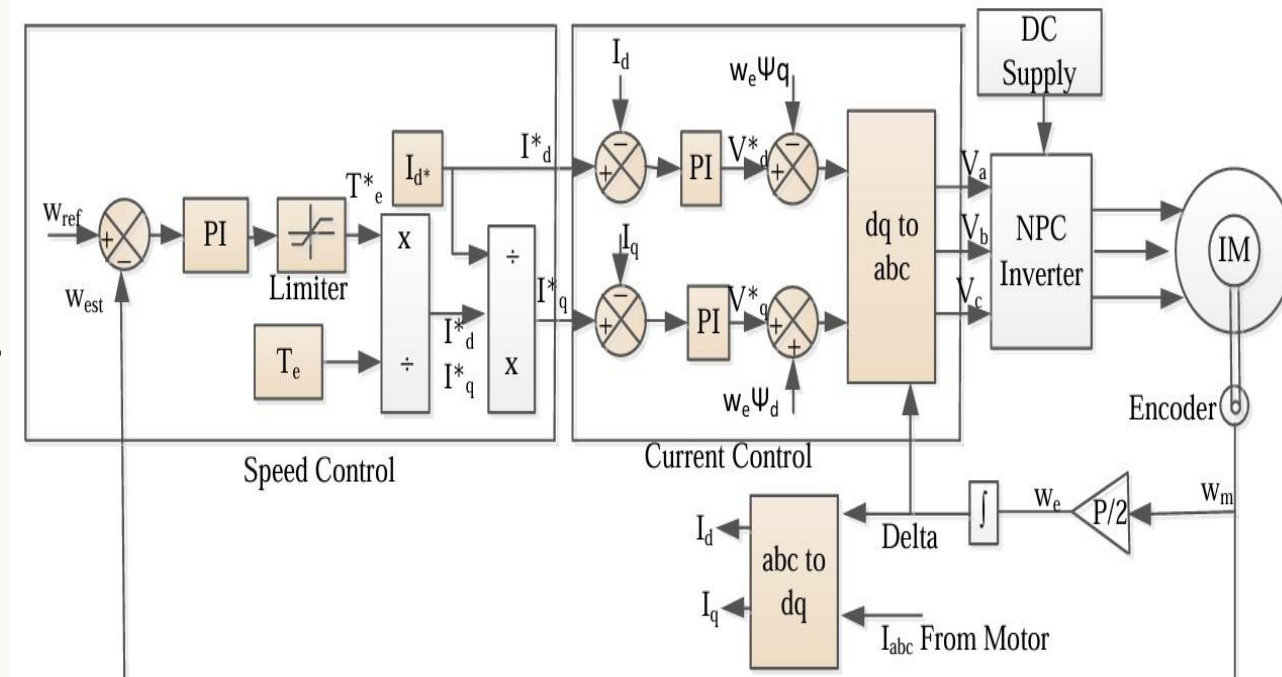
- Measured data is used as **feedback for current and speed control loops**.
- The **error between reference and measured values** is calculated.
- Errors are processed using **digital PI controllers**.
- The control algorithm is implemented in **Code Composer Studio**.

3. PWM Generation

- The controller generates **PWM switching pulses**.
- These pulses are amplified using **opto-coupler based gate drivers**.
- Gate drivers **activate inverter switches**.
- The inverter then supplies **regulated voltage to the induction motor**.

Advantages of FOC

- **Fast Response**
- **Independent Control**
- **High Efficiency**



Vector Control Schematic of Induction Motor

Vector Control Architecture

1. **Vector Control (FOC)** provides good dynamic performance by independently controlling **torque and flux**.
2. The system uses a **dual-loop control structure**, consisting of a **fast inner current loop** and a **slower outer speed loop**.
3. The **outer speed loop** generates the torque reference used to calculate the **q-axis current (I_q)**.
4. **Torque is controlled by the q-axis current**, while **flux is controlled by the d-axis current**, enabling independent control of speed and flux.

Vector Control Performance

1. **Vector Control (FOC)** provides good dynamic response by independently controlling **speed and flux**.
2. The system uses a **dual-loop control structure**: a fast **inner current loop** and a slower **outer speed loop**.
 - A. **Speed Control Loop**: Uses a **PI controller** to generate torque reference from speed error.
 - B. **Current Control Loop**: Calculates **d-axis and q-axis reference currents** for flux and torque control.
 - C. **Coordinate Transformation**: Converts **dq reference voltages to abc coordinates** for the NPC inverter.
 - D. **Motor Drive**: The **NPC inverter drives the induction motor** with encoder feedback.

Simulation Results: Step Load Change

Test Parameters

Motor Rating: 1.1 kW

Rated Torque: 7.3456 N·m

Rated Current: 2.55 A

Rated Speed: 1430 rpm

Stator Resistance (Rs): 5.1 Ω

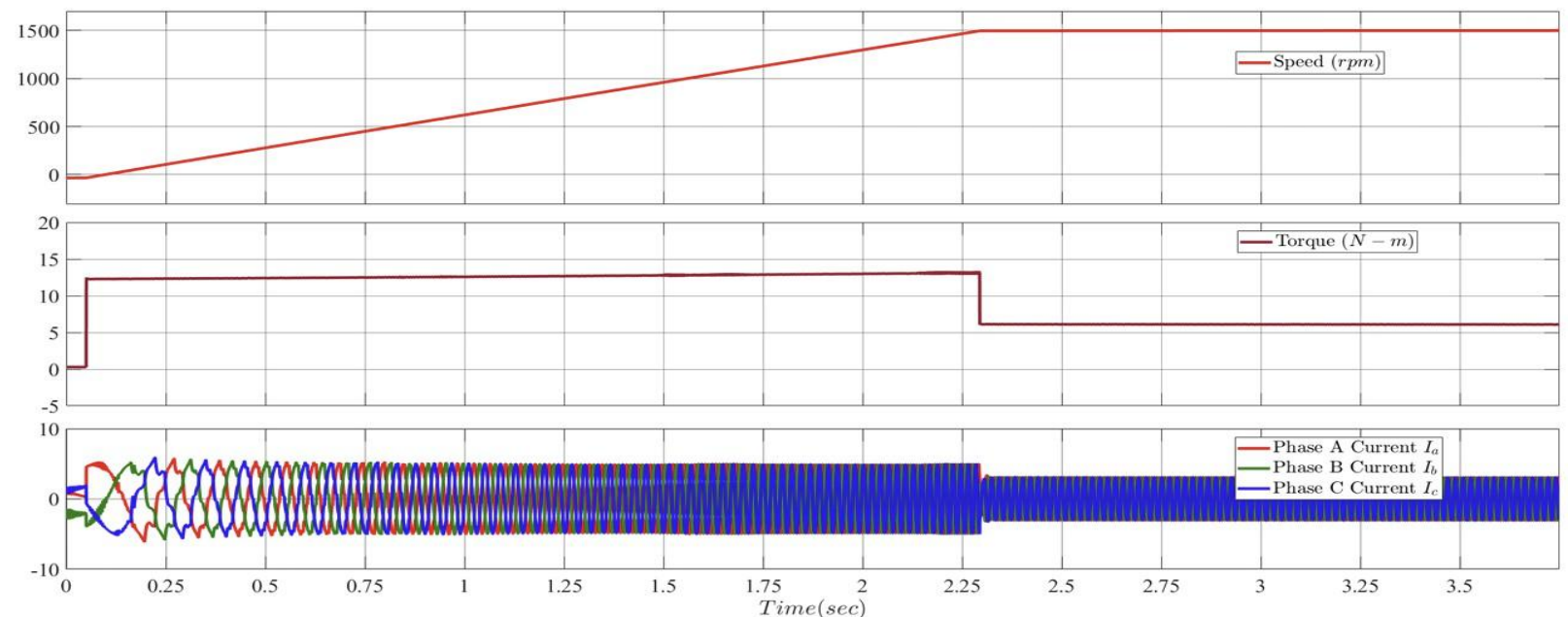
Rotor Resistance (Rr): 4.9 Ω

Mutual Inductance (Lm): 0.44189 H

Leakage Inductance (Llr): 0.0217 H

Dynamic Response

- The simulation shows **good dynamic performance** during step load changes.
- Motor speed increases **linearly up to 1500 rpm** with very small overshoot.
- Torque changes from **0 to 12 N·m at 2.25 s** and then reduce to **5 N·m**, indicating fast response.
- **Three-phase currents remain sinusoidal**, with amplitude varying according to torque changes.
- This confirms the **effective implementation of vector control**.



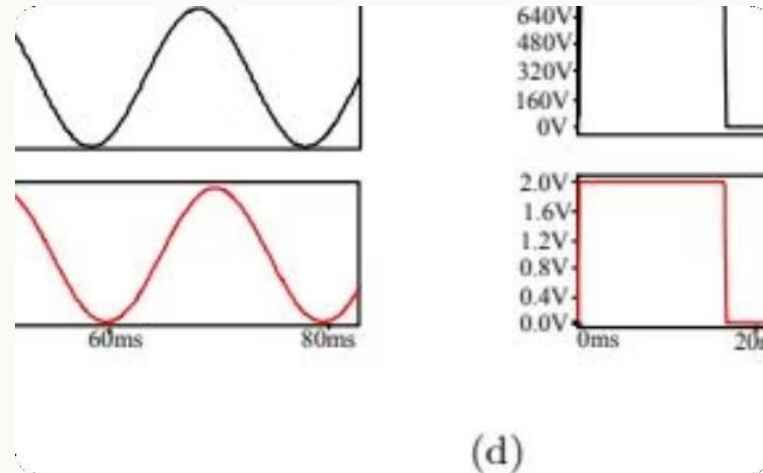
Simulation Result for Step Load Change

Hardware Validation Results



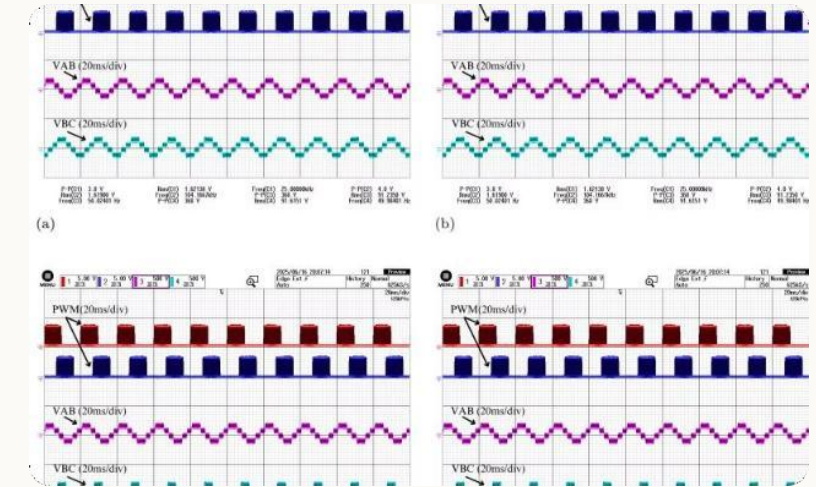
Hardware Prototype

Complete NPC converter with SiC MOSFETs, DC-link capacitors, gate drivers, and sensor circuits on professional PCB layout.



Sensor Validation

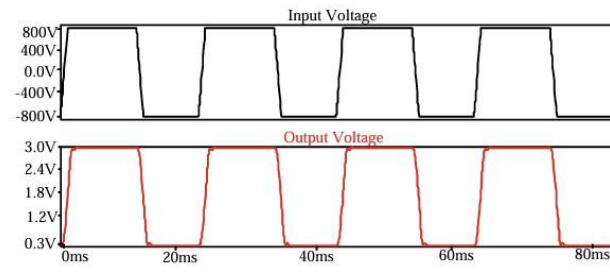
LTspice simulation confirms AC voltage, current, DC voltage sensors and gate driver circuits operate within microcontroller range.



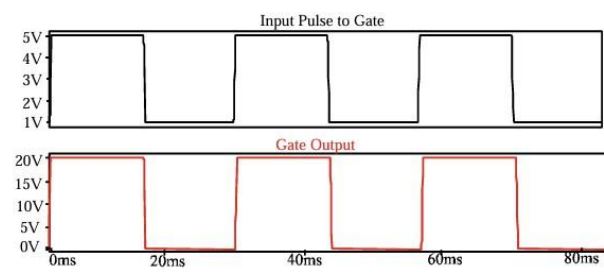
Experimental Waveforms

Oscilloscope captures show PWM pulses and multi-level line voltages VAB and VBC validating proper NPC operation.

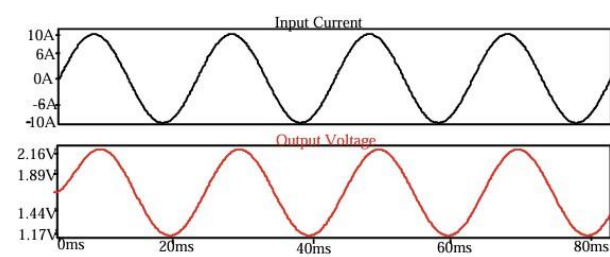
The developed system successfully demonstrates vector control of induction motor using SiC MOSFET based NPC converter with excellent dynamic response and clean multi-level voltage waveforms.



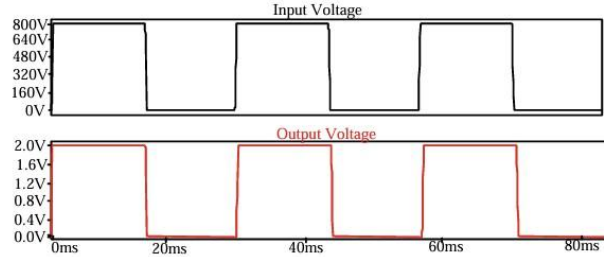
(a)



(b)

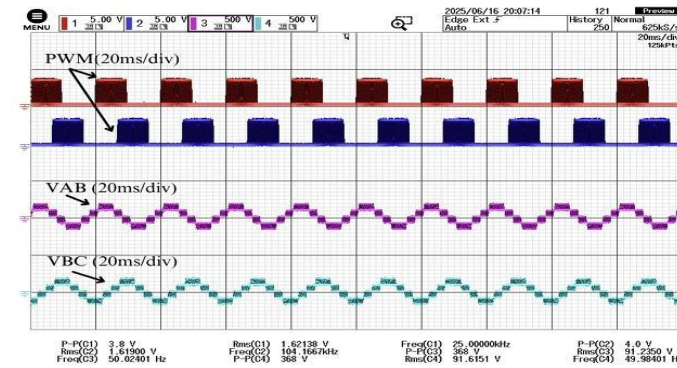


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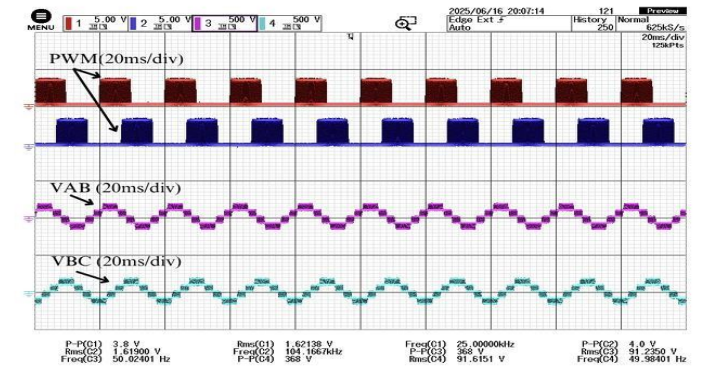


(d)

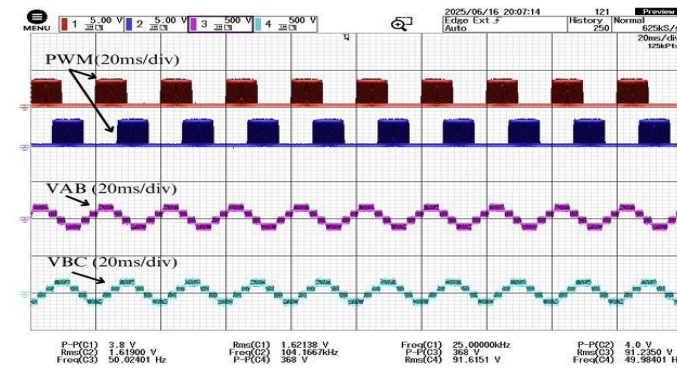
(a) AC Voltage Sensor (b) Gate Driver (c) Current Sensor (d) DC Voltage Sensor



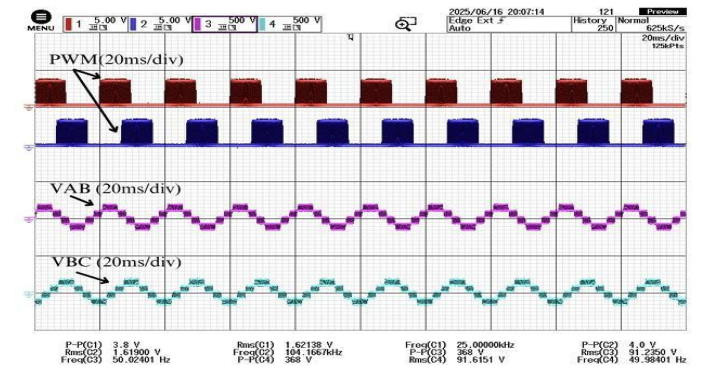
(a)



(b)



(c)



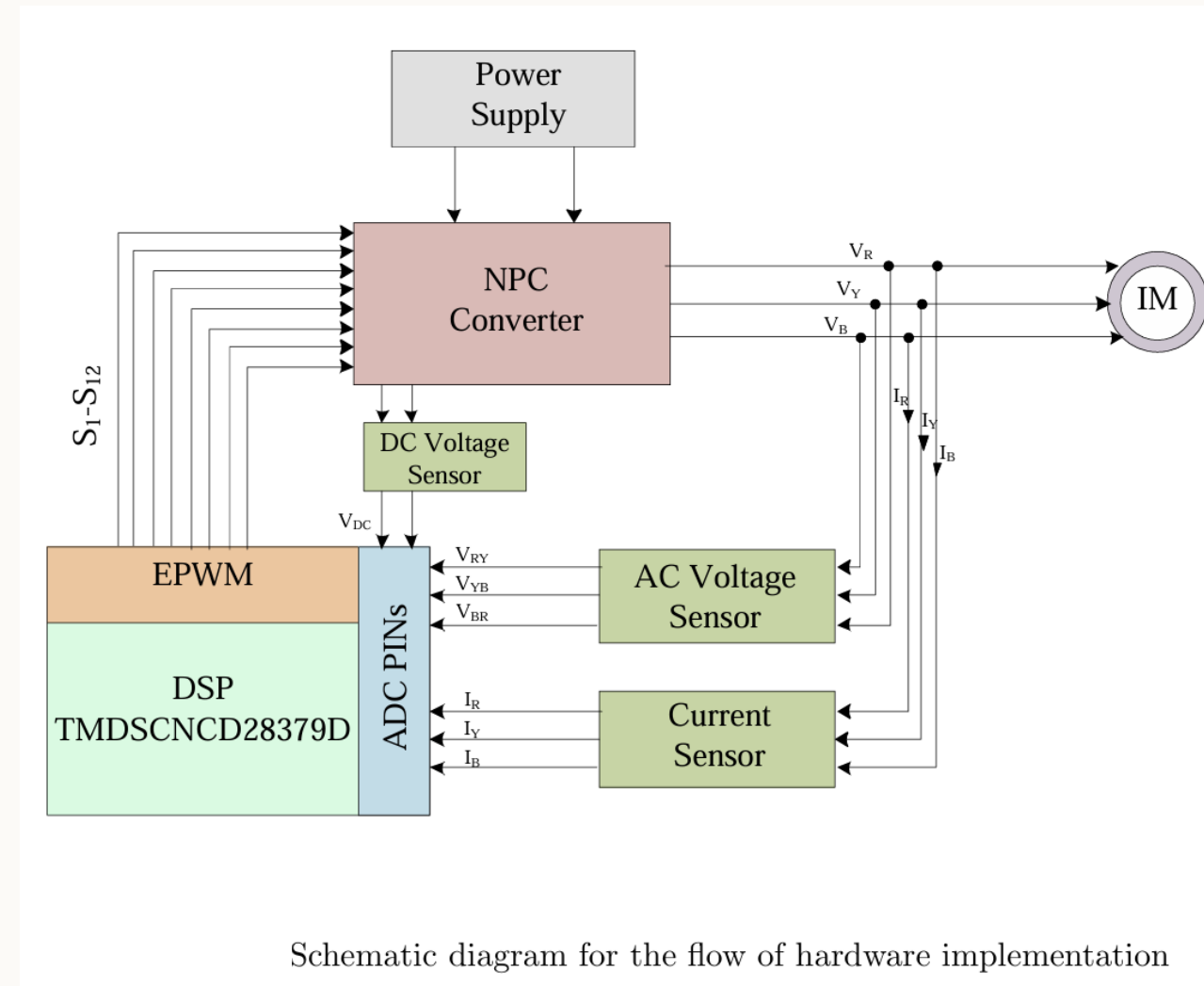
(d)

(a) AC Voltage Sensor (b) Gate Driver (c) Current Sensor (d) DC Voltage Sensor

The three-level NPC topology generates pole and phase voltages with 3 levels and line voltages with 5 levels. RMS line voltage is calculated as: $0.612 \times m \times (N-1) \times V_{dc}$, where m is modulation index and N is number of levels.

Assembly and Testing of SiC MOSFET NPC Converter

This chapter presents the hardware implementation of a three-level Neutral Point Clamped (NPC) converter using SiC MOSFETs. The developed prototype was tested with resistive loads and a 3-phase induction motor under both open and closed loop conditions.



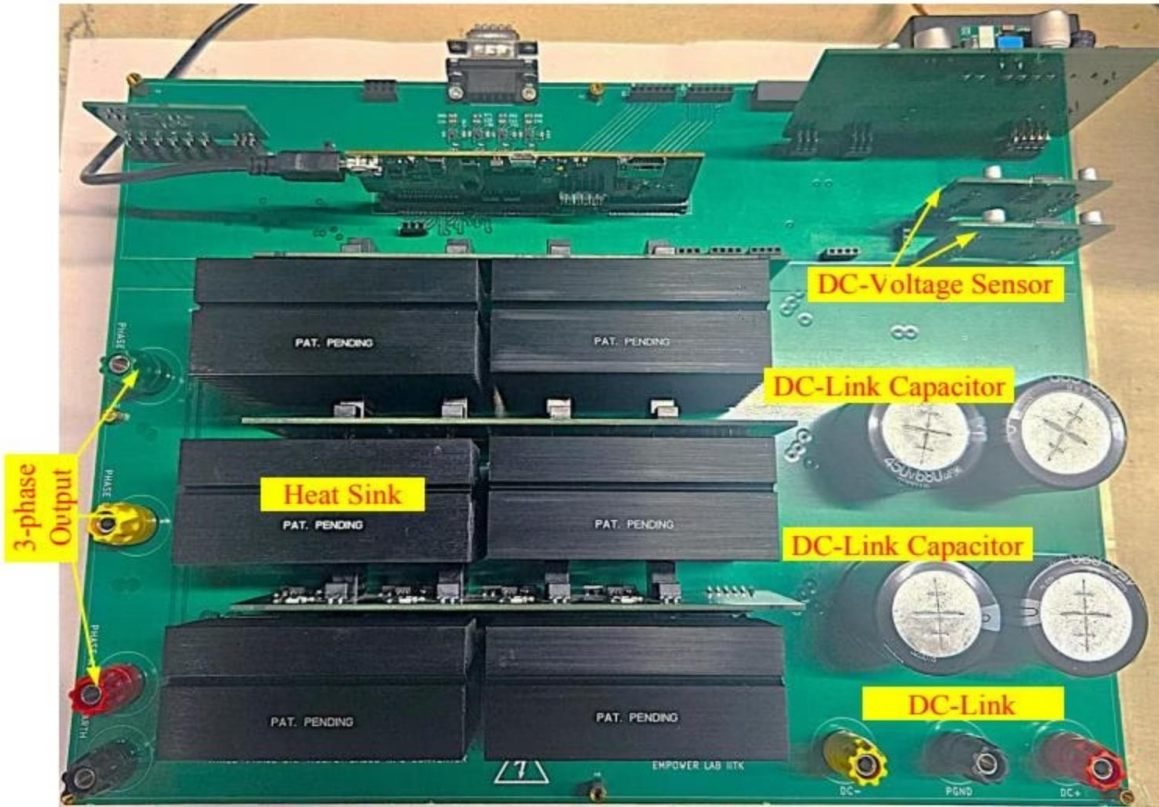
NPC Converter Design and Components

Design Parameters

- Output Power: 6.6 kW
- DC Link Voltage: 700 V
- AC Line Voltage: 415 V
- Max Frequency: 100 kHz

Key Components

- Gate Driver Circuit
- Heat Sink Assembly
- Voltage & Current Sensors
- Auxiliary Power Supply
- Snubber Circuit



Hardware Testing and Validation

AC Voltage & Gate Driver

Sensor output scaled to microcontroller range; gate driver amplifies PWM pulses to 20V for switch activation

1.1kW

Motor Power

Rated at 1430 rpm with
7.35 N-m torque

2.55A

Rated Current

Three-phase
operation

Current & DC Voltage Sensors

Current sensor handles $\pm 10\text{A}$ range; DC voltage sensor monitors 0-800V DC link

5.1 Ω

Stator Resistance

Rotor resistance:
4.9 Ω

Advantages, Limitations And Application of NPC

Key Merits

1. Reduced Component Requirement

The number of components such as **capacitors and switches** is lower compared to **FC and CHB multilevel inverters** for the same power rating.

2. Cost Effectiveness

The overall **system cost is lower** than FC and CHB MLI because **fewer passive components, especially capacitors, are required**.

3. High Efficiency

Optimized switching strategies help achieve **high conversion efficiency**.

4. Moderate Complexity

The system has **moderate design and control complexity** compared to **FC and CHB multilevel inverter topologies**.

Design Challenges

1. Scalability

- Diode count and circuit complexity **increase for voltage levels above 5**.
- Higher levels make the **NPC topology more complex to implement**.

2. Modularity

- **System expansion is difficult** in NPC converters.
- Unlike **CHB multilevel inverters**, NPC is **not naturally modular**.

3. Neutral Point Balancing

- **Unbalanced operation can cause voltage drift** at the neutral point.
- Additional **control techniques may be required** to maintain voltage balance.

Industrial Applications

1. Motor and Traction Drives

- NPC provides a good **balance between efficiency, cost, and complexity**.
- Widely used in **railway traction motor drives** to handle **high power demands**.

2. EV Charging Applications

- NPC MLI is suitable for **electric vehicle charging systems**.
- Offers **high efficiency, better waveform quality, and fault tolerance capability**.

3. Other Applications

- Used in **HVDC transmission systems**.
- Applied in **renewable energy systems** such as **solar PV and wind turbines**.
- Also used in **UPS (Uninterruptible Power Supply) systems**.

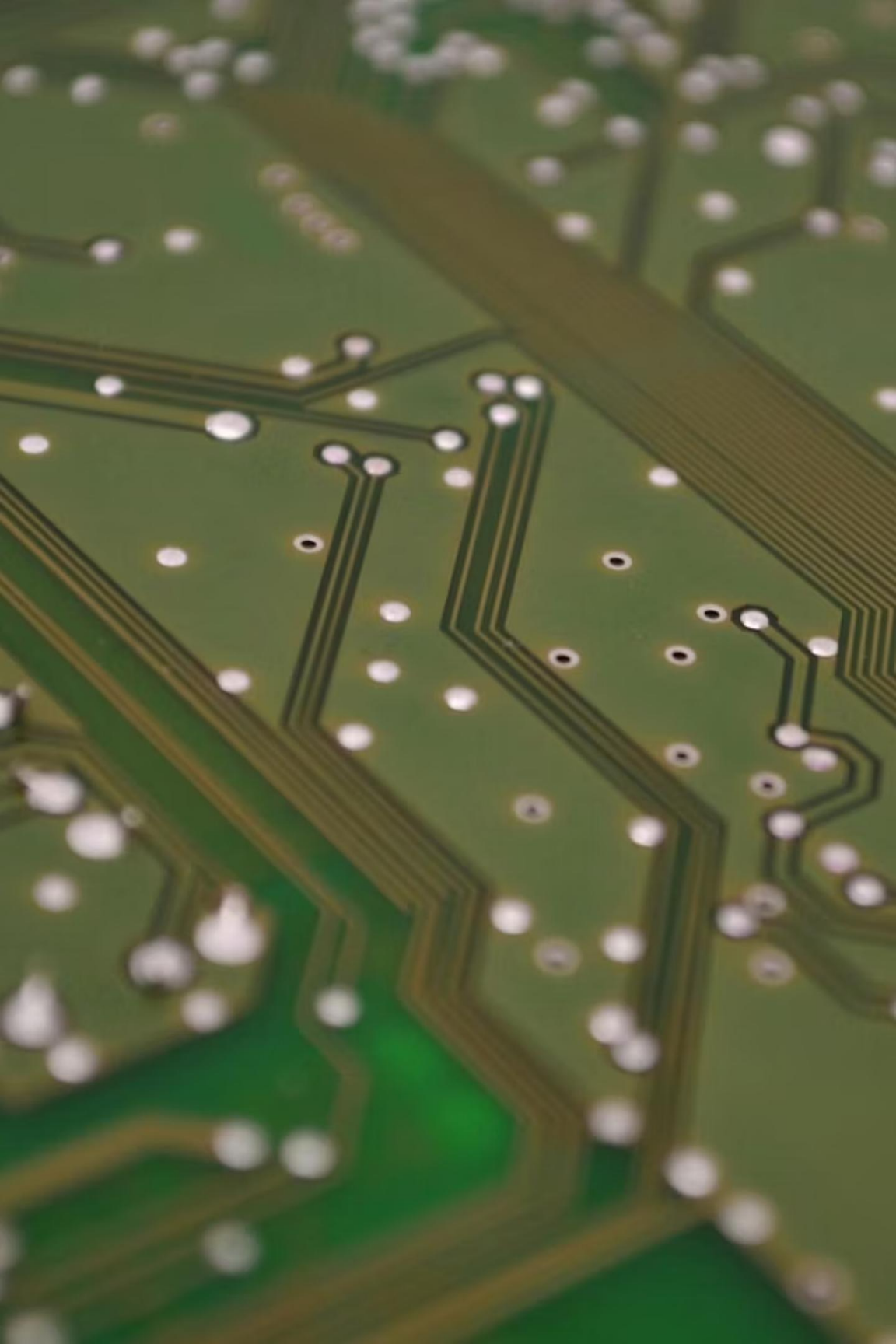
Key Achievements and Future Directions

Main Conclusions

- A **compact and robust 3-level NPC converter** was successfully designed and implemented for **medium-power applications**.
- The system was validated using **MATLAB/Simulink simulations**.
- **Hardware testing** was performed to verify the practical operation of the converter.
- The control strategy used **Vector Control (FOC)** along with **Common Mode Voltage Injection (CMVI)**.

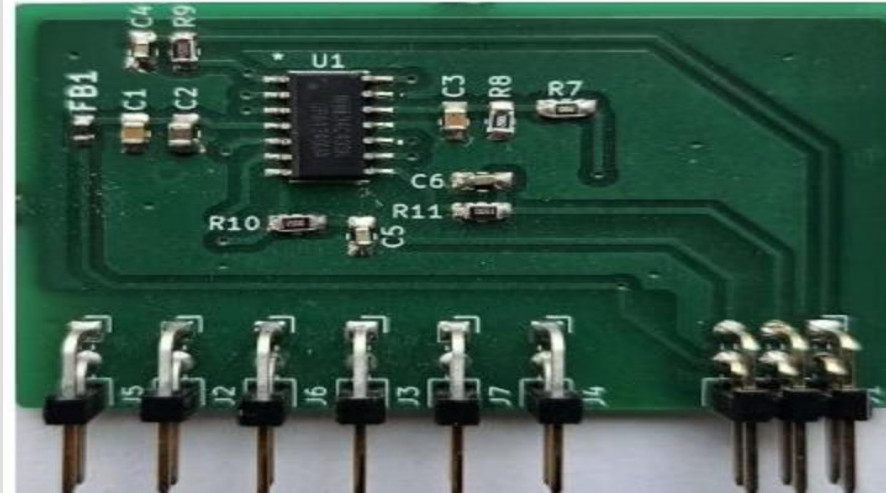
Future Scope of NPC

- Future work aims to develop **NPC converters for medium and high-power applications** with improved efficiency.
- Efforts will focus on **better neutral point voltage balancing** and **reducing the number of components**.
- Advances in semiconductor technology such as **Silicon Carbide (SiC)** and **Gallium Nitride (GaN)** will enhance system performance.
- These technologies will help achieve **lower switching losses and more compact converter designs**.
- With the increasing demand for **electric vehicles, drive systems, and grid applications**, NPC converters will play an important role.
- NPC topology will continue to be significant for **energy-efficient and high-power applications** in future power electronics systems.



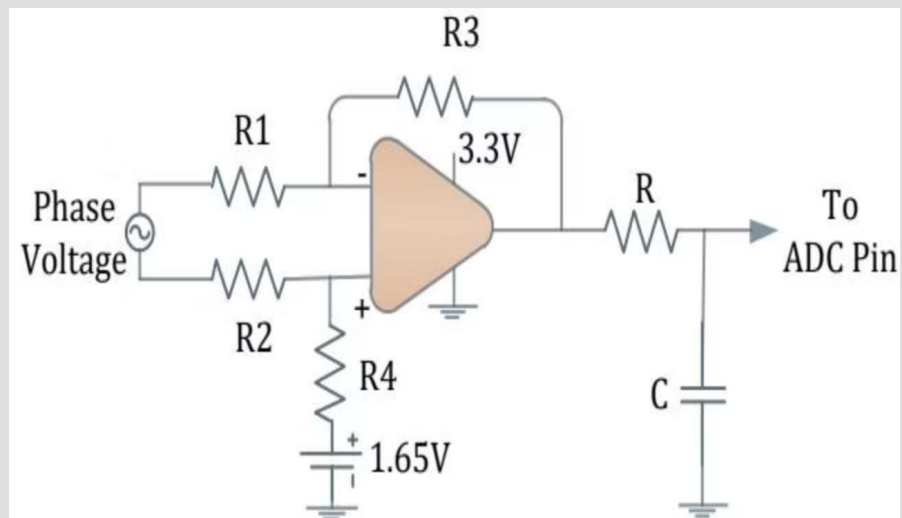
Voltage Sensing Systems

This presentation details the schematic design, component selection, simulation results, and practical implementation considerations for isolated voltage sensing in high-power applications.



AC Voltage Sensing Circuit Design

A report to three-phase AC voltage sensing for power electronics converter applications



AC Voltage Sensing Circuit

- The **AC voltage sensor is an important component** in power electronics converter systems.
- It **measures the output AC voltage** and provides **feedback in closed-loop control**.
- This feedback helps maintain **stable and efficient system operation**.
- The selection of an AC voltage sensor depends on **voltage level, accuracy, and response time**.
- In motor drive systems, if the converter output is **not close to a sinusoidal waveform**, the motor may not run smoothly.
- The sensor detects such variations and **sends feedback to maintain stable motor operation**.
- The designed sensor measures **three-phase AC voltage using a resistor divider and an operational amplifier**.
- The circuit converts the sensed voltage into a **signal compatible with the microcontroller (MCU)**.

System Stability

Provides essential feedback for closed-loop control systems

Motor Control

Ensures sinusoidal output for smooth motor operation

Efficiency

Enables robust and efficient converter design

Designed AC Voltage Sensor

- The designed **AC voltage sensor uses a resistor divider circuit and an operational amplifier (Op-amp)** to measure the voltage.
- A **single two-layer PCB** is used to sense the **three-phase AC voltage**.
- The **Op-amp operates with a 3.3 V DC supply**, which is provided by the auxiliary power supply.
- A **1.65 V offset** is introduced so that the output voltage remains within the **0–3.3 V range of the microcontroller (MCU)**.
- A **differential amplifier configuration** is used since the phase voltage is measured **without a direct ground reference**.
- The **phase voltage is first reduced using a resistor divider network**, where **$R1 = R3$ and $R2 = R4$** , and the sensor gain is determined by **$R3/R1$** .
- The **high-voltage section is placed on the main PCB**, while the **low-voltage sensing circuit is located on the AC voltage sensor PCB**.
- An **RC filter and ferrite bead** are used to reduce noise, and the filtered output is sent to the **MCU as a 0–3.3 V signal**.

Phase Voltage Input Three-phase AC voltage fed through resistor divider circuit 04	Resistor Divider Steps down high voltage to manageable levels 05	Differential Amplifier Processes phase voltage without ground reference
RC Filtering Minimizes noise at output stage	MCU Interface 0-3.3V output compatible with microcontroller	

Circuit Design Specifications

Key Components

- The **operational amplifier requires a 3.3 V DC supply** for proper operation.
- This supply is provided by an **auxiliary power supply** that converts **230 V AC to 3.3 V and 5 V DC**.
- The generated DC voltages are used to power the **microcontroller (MCU), gate driver, and sensor circuits**.
- An **RC filter is placed at the output of the operational amplifier** to reduce noise in the output signal.
- The filtered signal is sent to the **MCU within the voltage range of 0–3.3 V**.
- A **ferrite bead (FB)** is connected with the operational amplifier to **reduce input noise**.
- The circuit is implemented on a **two-layer PCB designed using KiCad software**.
- The **high-voltage circuit is placed on the main board PCB**, while the **low-voltage circuit is placed on the AC voltage sensor PCB**.

Differential Amplifier

Used because we are sensing phase voltage without ground reference

1.65V Offset

Center point for 0-3.3V MCU-compatible output range

Two-Layer PCB

Designed in KiCad software for compact layout

AC Voltage Sensor Schematic



AC Phase Voltage

Input source connected through R1 and R2



Resistor Divider

R1, R2, R3, R4 form voltage scaling network



Op-Amp Processing

Differential amplifier with 1.65V bias



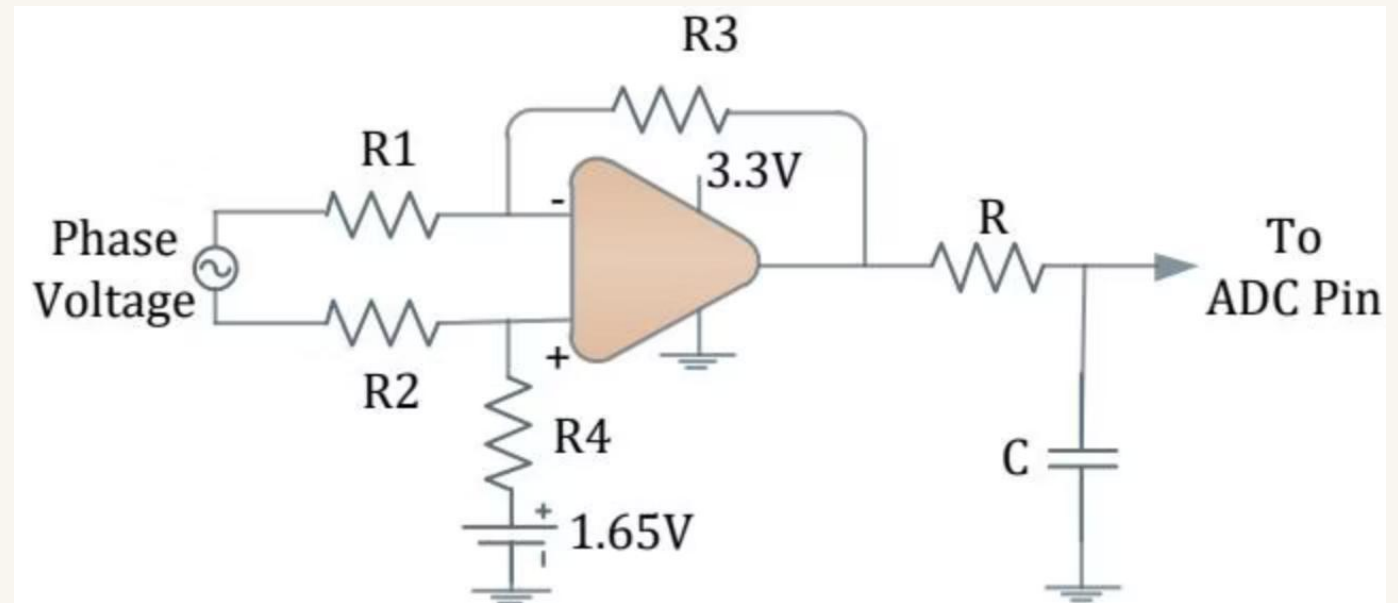
RC Filter

Series R and shunt C to ground



ADC Interface

0-3.3V output to MCU



- The **non-inverting input (+)** of the op-amp is connected to a **1.65 V DC bias** through resistor **R4**.
- The **inverting input (-)** is connected to the **output via feedback resistor R3**.
- The **gain of the sensor** is determined by the ratio **R_3 / R_1** .
- The resistor values are selected such that **$R_1 = R_3$** and **$R_2 = R_4$** .
- This configuration ensures **proper signal scaling and stable sensor operation**.

Selected Components and Values

S. No	Component	Value/IC Name
1	Op-Amp	AD8086
2	Ferrite bead	26 Ω
3	RC Filter	R=150 Ω , C=1nF
4	Size of PCB	50mm x 36mm
5	R1, R2	3001k Ω
6	R3, R4	5k Ω

- **Operational Amplifier**
The **AD8086 operational amplifier** is chosen for its **differential amplification capability**, making it suitable for sensing phase voltage **without requiring a ground reference**.
- **RC Filter Network**
A combination of **R = 150 Ω** and **C = 1 nF** forms an **RC filter** that reduces noise at the output stage and ensures a **clean signal is delivered to the MCU**.
- **Resistor Divider**
High-value resistors (3001 k Ω and 5 k Ω) are used to **scale down the input voltage** to an appropriate level while maintaining **high input impedance**.

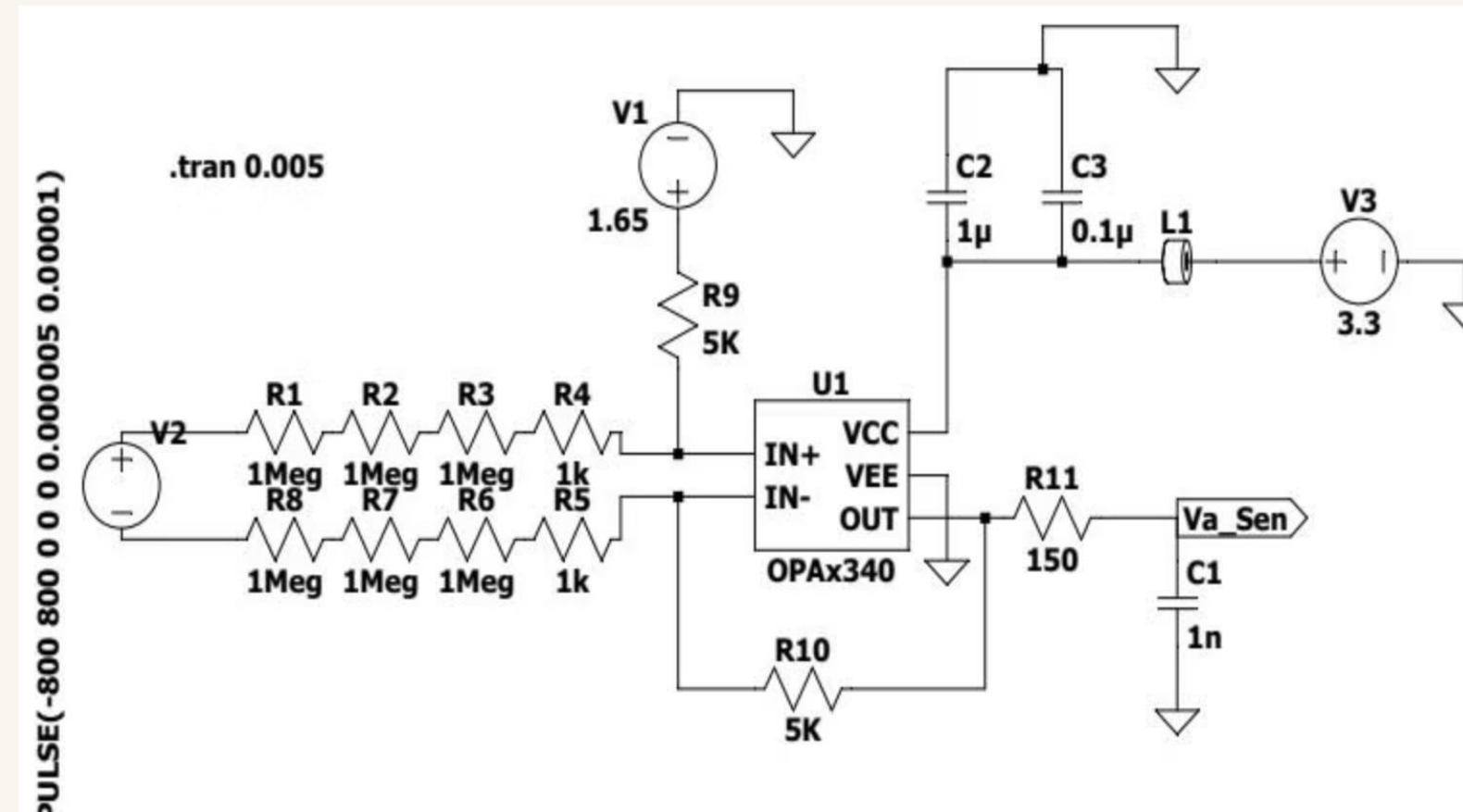
LTspice Simulation Results

- The designed **AC voltage sensor** was first simulated using **LTspice software** to analyze its performance.
- The **gain of the sensor circuit** was determined through simulation results.
- The sensed output voltage is calculated using the formula:

$$V_{out} = (V_{in} - 1.65) \times 600$$

- In this equation, **1.65 V** acts as the **offset voltage** and **600** represents the **gain of the circuit**.
- The offset value is later **adjusted during calibration with actual hardware measurements**.
- The simulation uses a **pulse source (V2)** defined as PULSE(-800 800 0 0 0 0.000005 0.000001) with the simulation command **.tran 0.005**.

- The input signal is connected through a **series network of resistors R1–R8**, where **R1–R4** are **1 MΩ** and **R5–R8** are **1 kΩ**.
- The **junction of R4 and R5** is connected to the **non-inverting input (IN+)** of the operational amplifier **U1 (OPAx340)**.
- The **inverting input (IN-)** is connected to a **1.65 V** reference source through resistor **R9 (5 kΩ)** and to the **output** through resistor **R10 (5 kΩ)**, forming a **unity-gain feedback loop**.



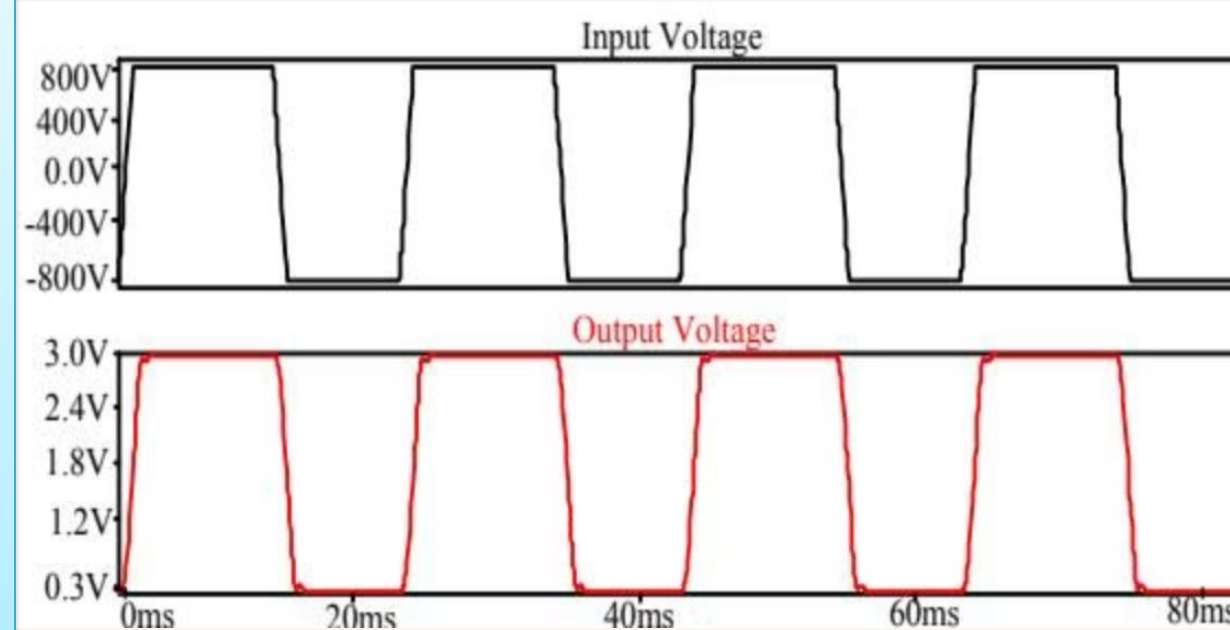
Simulation Waveform Analysis

Input Voltage

- The **input voltage waveform** is a square wave varying between **+800 V** and **-800 V**.
- It oscillates with a **time period of about 80 ms**.
- This signal represents the **three-phase AC voltage that must be sensed and scaled** before sending it to the microcontroller.

Output Voltage

- The **output voltage waveform** is a square wave ranging approximately between **0.3 V** and **3.0 V**.
- This range makes the signal **compatible with the MCU input range**.
- The output waveform is **phase-inverted** compared to the input, which is typical of an **inverting amplifier configuration**.



- The **output of the operational amplifier** is connected to a **low-pass filter network**.
- The filter consists of **resistor R11 (150 Ω)**, **capacitor C2 (1 μ F)**, **capacitor C3 (0.1 μ F)**, and **inductor L1**.
- The filtered output signal is labeled **“Va_Sen”**.
- **Capacitor C1 (1 nF)** is connected to ground to provide **bypass filtering and reduce noise**.
- A **DC voltage source V3 (3.3 V)** is also present at the output stage.



Time Period
80ms simulation duration



Output Range
0.3V to 3.0V compatible

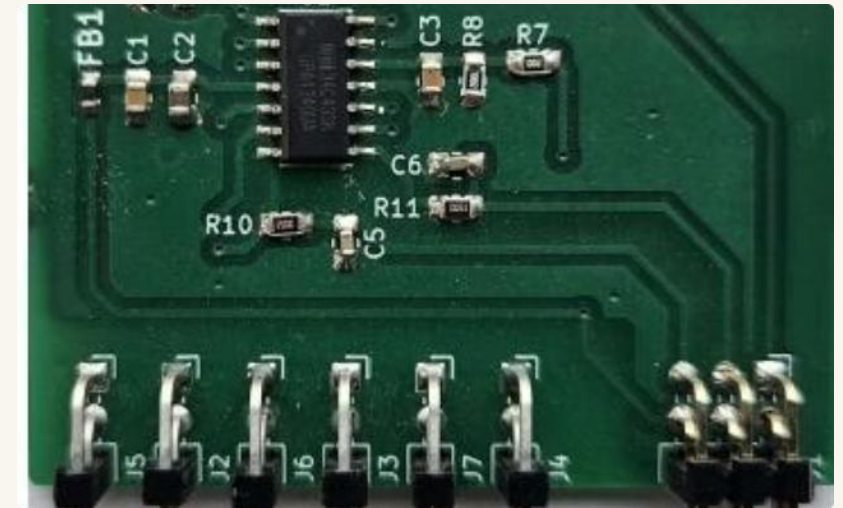
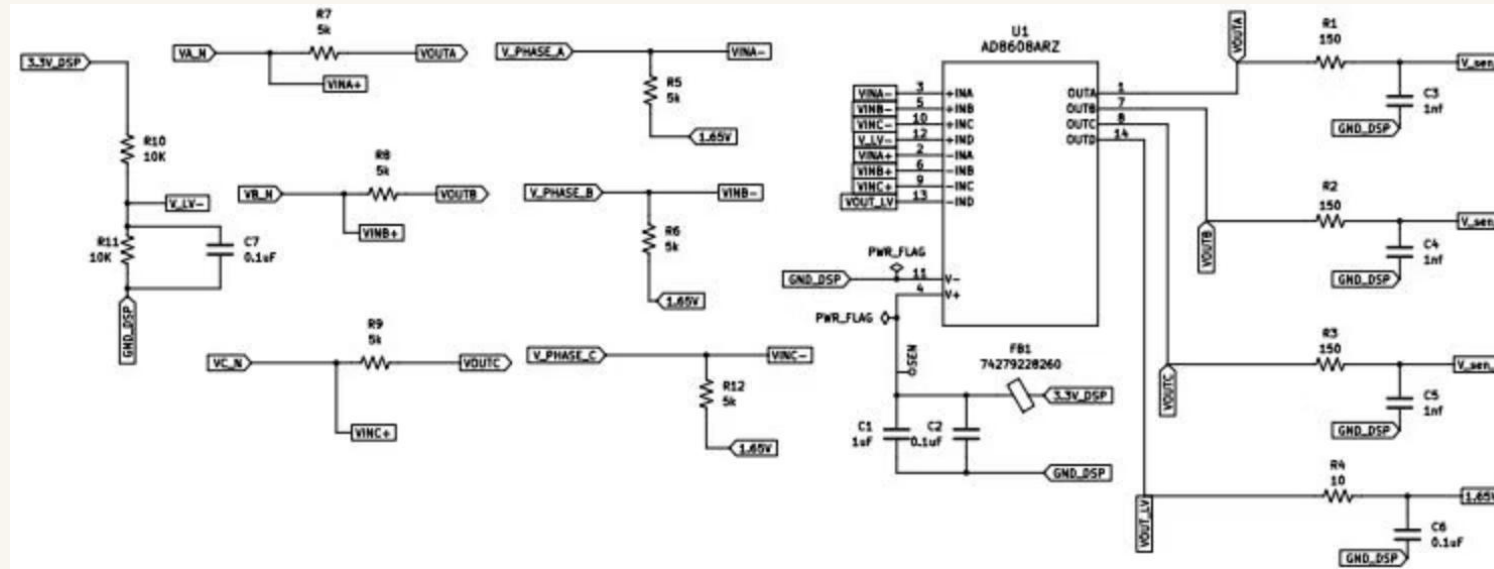


Input Range
 ± 800 V peak-to-peak



Phase Inversion
180° phase shift observed

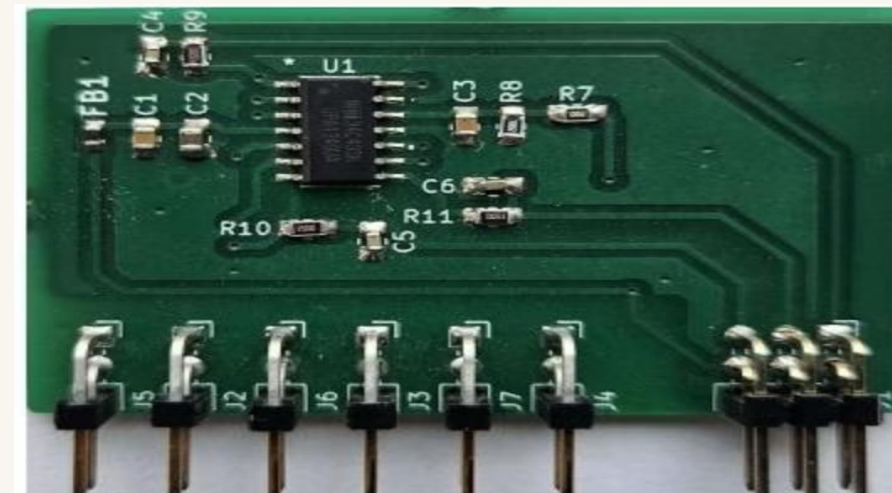
KiCad Schematic and PCB Design

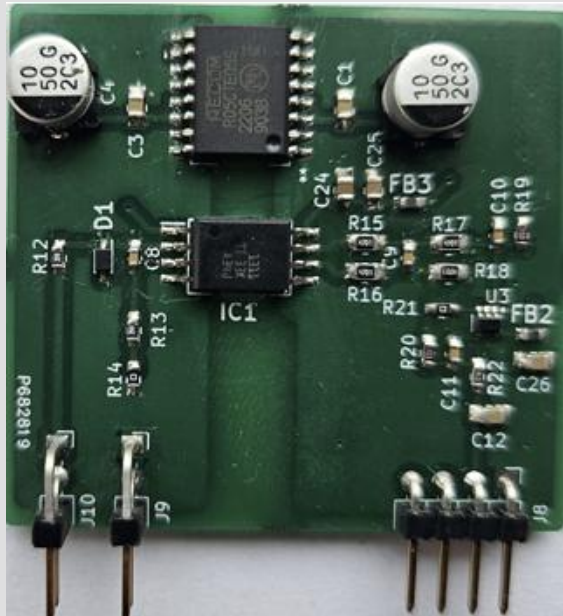
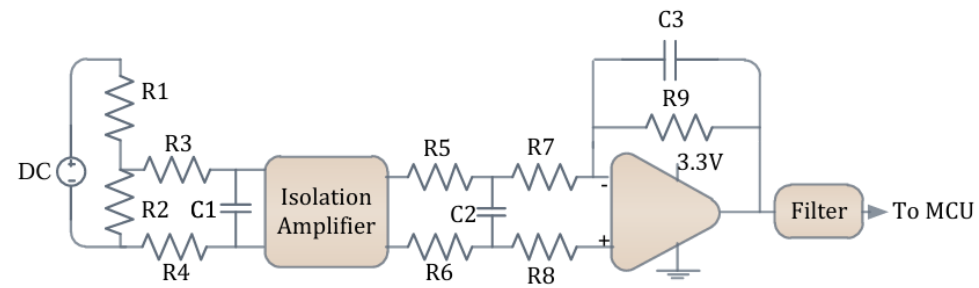


- The **AC voltage sensor schematic and PCB** were designed using **KiCad software**.
- The schematic represents a **three-phase AC voltage sensing circuit**.
- The **input section includes three AC voltage sources (V1, V2, V3)** connected through a network of **resistors and capacitors**.
- These signals are applied to the **non-inverting inputs of the operational amplifier** for signal conditioning.
- The PCB is a **two-layer board with dimensions of approximately 50 mm × 36 mm**.
- Components such as **resistors, capacitors, and connectors** are **arranged carefully** to maintain good signal integrity.
- The **operational amplifier outputs are connected to LED indicators** through current-limiting resistors and capacitors.
- Additional **passive components** are included for **filtering, biasing, and stable circuit operation**.

Physical PCB Design of AC Voltage Sensor

1. The **designed AC voltage sensor PCB** is rectangular in shape with dimensions of **50 mm × 36 mm**.
2. The board contains a **central integrated circuit (U1)** along with passive components such as **resistors and capacitors**.
3. **Two multi-pin connectors** are provided at the bottom for electrical connections.
4. The **compact PCB size** allows easy integration into **space-constrained designs**.
5. The design uses a **two-layer PCB** to enable efficient routing and separation of high-voltage and low voltage sections.
6. **High-voltage circuits** are placed on the **main board PCB** for better safety and isolation.
7. **Low-voltage circuits** are placed on the **AC voltage sensor PCB** for interfacing with the **microcontroller (MCU)**.
8. The system includes an **auxiliary power supply** that converts **230 V AC into 3.3 V and 5 V DC**.
9. **Ferrite beads and RC filters** are used to reduce noise and improve signal quality at the operational amplifier output.
10. The filtered output is sent to the **MCU in the range of 0–3.3 V**, ensuring compatibility with microcontroller inputs.





DC Voltage Sensing Circuit Design

A report to DC voltage sensing for power electronics converter applications

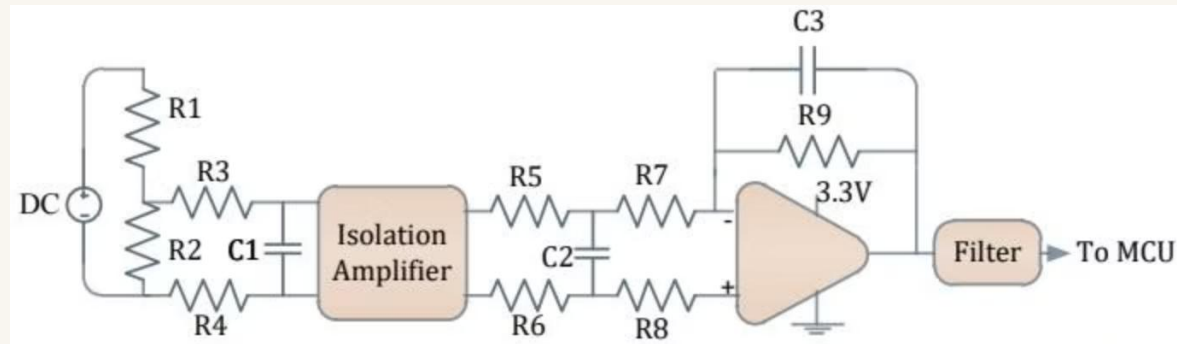
DC-Link Voltage Sensor

- The DC-link voltage is reduced using a **resistor voltage divider** to a safe range (0–2 V) for the isolation amplifier.
- An **isolation amplifier (AMC1311)** is used to provide electrical isolation and protect low-voltage circuits.
- The isolated output is passed to an **op-amp for filtering and signal conditioning**.
- A **second op-amp** is used to amplify the signal to the **MCU compatible range (0–3.3 V)**.
- A **low-pass RC filter** is placed between stages to remove noise from the signal.
- The system uses an **auxiliary power supply (3.3–5 V)** to operate the isolation amplifier and op-amps.
- **Two op-amps are used**: one for isolation/conditioning and the other for gain adjustment.
- Resistor values (**R1, R2**) are calculated to scale high DC voltage ($\approx 700\text{--}800\text{ V}$) to a safe input ($\sim 1.15\text{ V}$).
- The **filter cutoff frequency** is selected as about **1/10th of the switching frequency** (e.g., $R = 100\ \Omega$, $C = 0.1\ \mu\text{F}$).
- The final output is a **stable, noise-free voltage signal (0–3.3 V)** suitable for microcontroller measurement and control.

Features of DC-Link Voltage Sensor

- **Overvoltage Protection**: Continuously monitors DC-link voltage and triggers protection when it exceeds safe limits to prevent damage.
- **Undervoltage Detection**: Detects low input voltage conditions that may cause unstable or faulty operation.
- **Output Regulation**: Maintains a stable output voltage using closed-loop feedback despite input variations.
- **Transient Response**: Responds quickly to sudden voltage spikes or changes for immediate correction.
- **System Importance**: Ensures safe, stable, and efficient converter operation by monitoring and regulating DC-link voltage.

DC Voltage Sensor Circuit Design



- The circuit uses a **voltage divider** to reduce high DC input voltage to a safe **0–2 V range** for the isolation amplifier.
- The **isolation amplifier** provides electrical isolation between high-voltage input and low-voltage output, operating with a **3.3–5 V supply**.

• The output of the isolation stage is sent to an **op-amp for filtering and signal conditioning**.

• An **RC low-pass filter** is placed between stages to minimize noise and interference, with cutoff frequency set lower than the switching frequency.

• The **second op-amp stage** amplifies the signal to a **0–3.3 V range**, making it suitable for MCU input.

Component Calculation

Resistor values calculated for 800V input to produce 1.15V at isolated amplifier input (safe operating point):

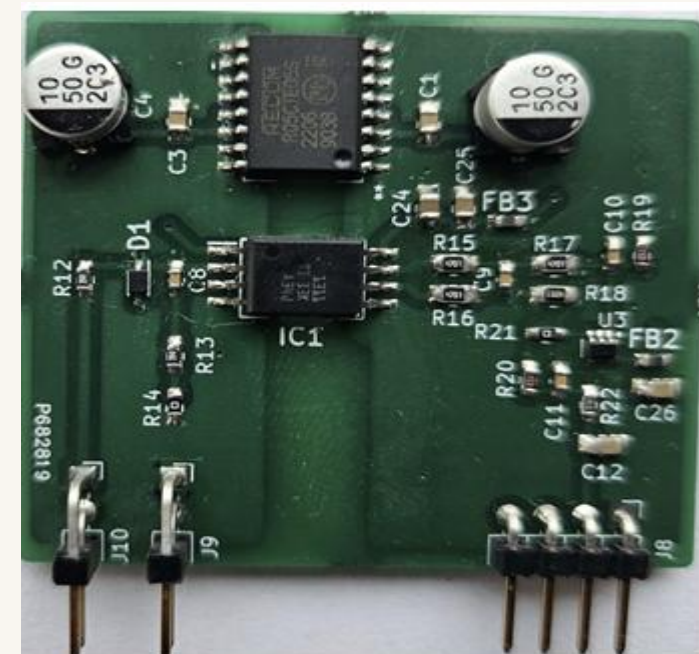
$$1.15 = \frac{V_{DC} \times R_2}{R_1 + R_2} \quad \Rightarrow \quad 1.15 = \frac{800 \times R_2}{(6 \times 332 \times 10^3) + R_2}$$

$$R_2 = 3.24K\Omega$$

RC filter values selected for cutoff frequency of 16 kHz:

$$f = \frac{1}{2\pi RC} \quad \Rightarrow \quad 16 \times 10^3 = \frac{1}{2\pi RC}$$

Resulting in: **R = 100Ω, C = 0.1μF**



LTspice Simulation: Schematic and Setup

Simulation Overview

- The DC voltage sensor circuit was initially **simulated in LTspice** before hardware implementation to verify its performance.
- The simulation includes key elements such as a **DC source (0–800 V pulse)**, **resistor divider**, **isolation amplifier (AMC1311)**, **two op-amps**, and **RC filters**.

Component Details

- The **resistor divider** is formed using six **332 k Ω resistors in series** to scale down the voltage.
- The **isolation amplifier (AMC1311)** uses a **0.01 μ F input capacitor** for stable operation.
- Two op-amps are used: a **generic op-amp for initial processing** and a **TLV9062 with output filtering**.
- A **final RC filter (100 Ω and 0.1 μ F)** is used to reduce noise and provide a clean output signal.

3 Stages are

1. Dual Sensor Setup:

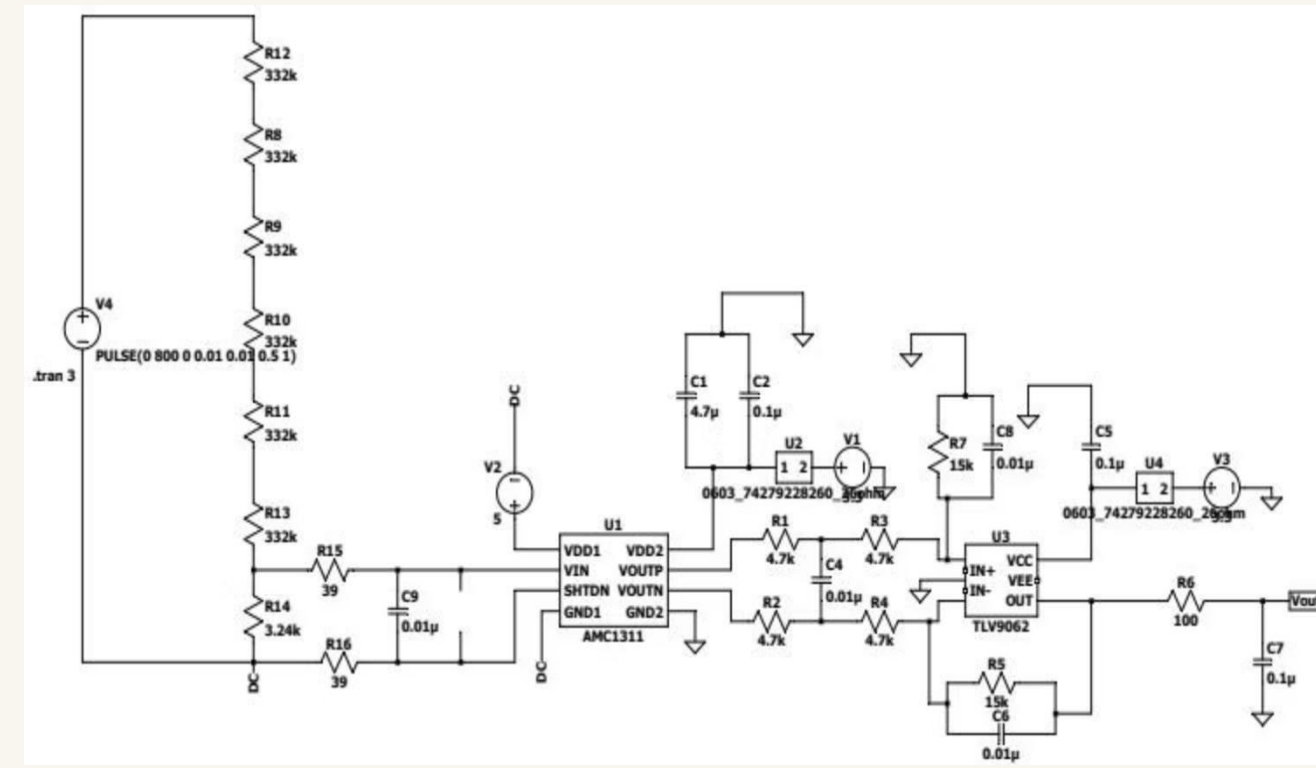
Two DC voltage sensors measure the top and bottom DC-link capacitor voltages to support **NPC balancing**.

2. Protection Circuit:

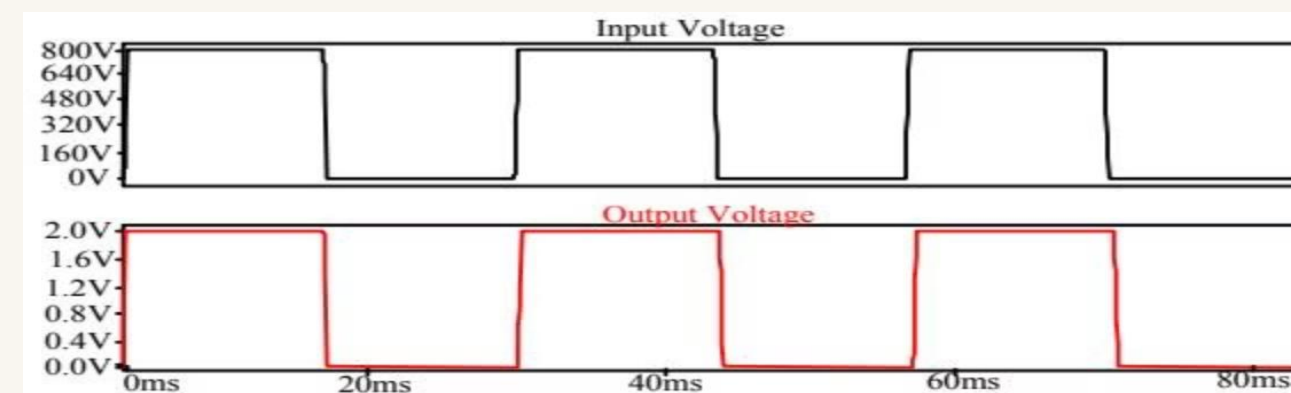
A **Zener diode (ZD1)** is used at the input to provide **overvoltage protection**.

3. Filtering Stage:

Multiple **RC filter networks** ensure a **clean and noise-free signal** at different stages of processing.



LTspice Simulation Results Analysis



- The simulation confirms the **correct functioning** of the DC voltage sensor circuit.
- The input voltage is a **square wave varying from 0 V to 800 V** with a period of 80 ms.
- The output voltage follows as a **scaled square wave (0–2 V)**, verifying proper voltage scaling and isolation.

800V

Maximum Input

Peak voltage applied to
sensor input

2.0 V

Output Range

Isolated amplifier output
compatible with MCU

80ms

Simulation Period

Total time duration of
simulation

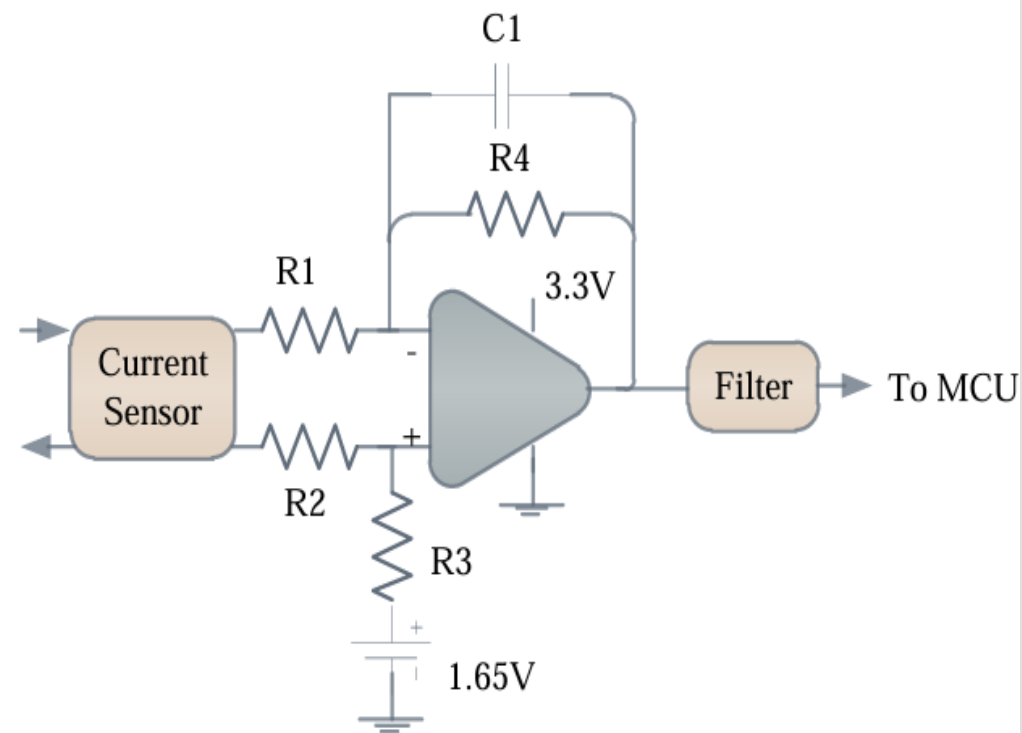
16kHz

Filter Cutoff

RC filter frequency for
noise rejection

Transient Behavior

- The output waveform shows **minor transients at signal edges**, which occur due to the RC filter networks.
- The filters effectively **reduce high-frequency noise** while preserving signal quality.
- The **0–2 V output range** is suitable for a **3.3 V microcontroller**, ensuring proper ADC operation.



Current Sensing Circuit Design

A report to Current sensing for power electronics converter applications.

A comprehensive study to hall effect-based current sensing circuits for three-phase power converter applications.

Current Sensing Circuit

- The **current sensor** is a key component in power converters for monitoring and controlling current to ensure stable and efficient operation.
- It provides **real-time feedback**, and its accuracy directly affects overall system performance.
- The sensor also offers **overcurrent protection**, preventing damage to the converter.
- A **Hall-effect based current sensor** is used, where the output voltage is proportional to the input current.
- It provides **electrical isolation** between power and control circuits, reducing noise and high-voltage interference.
- The sensor measures **three-phase currents** and sends feedback signals to the **MCU for closed-loop control**.
- The output of the sensor includes an **offset voltage (~1.65 V)** and gain, and is further processed using a **differential op-amp**.
- The op-amp adjusts the signal to the **0–3.3 V range**, making it compatible with the microcontroller.
- A **low-pass RC filter** is used at the output to remove noise before feeding the signal to the MCU.
- The selected sensor (**ACS730, 20 A rating, 1 MHz bandwidth**) ensures safe operation, with simulation confirming output remains within MCU limits.

Hall Effect Sensor Operation:

- **Current Flow:** When current passes through a conductor, it generates a magnetic field proportional to the current.
- **Magnetic Interaction:** This magnetic field interacts with a semiconductor placed perpendicular to the current path.
- **Voltage Generation:** Due to the Hall effect, a voltage is induced across the semiconductor, perpendicular to both current and magnetic field.
- **Output Relation:** The induced voltage is directly proportional to the input current, enabling accurate measurement.

Designed Current Sensor Circuit

- The designed current sensor uses **Hall-effect technology** to provide electrical isolation between power and control circuits, reducing voltage interference.
- It is capable of **monitoring the phase currents of all three phases simultaneously**.



Sensor IC

Connected in series with the phase line and provides an output voltage compatible with the microcontroller.



Differential Amplifier

Uses an op-amp to amplify the signal and adjust it to the MCU input range.



Low-Pass Filter

An RC filter smooths the signal before sending it to the microcontroller.

Output Voltage Equation

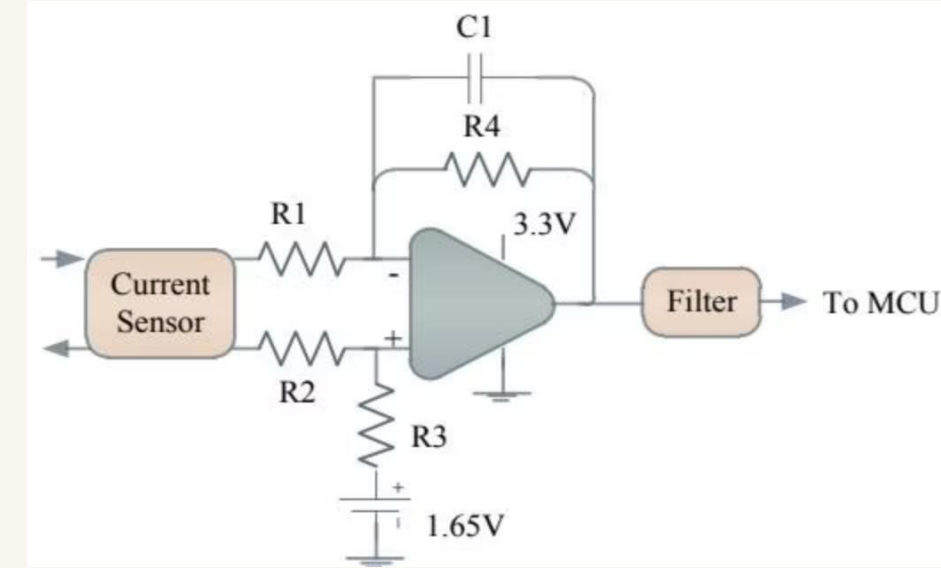
- The output voltage is proportional to input current with a fixed offset.

$$V_{OUT} = \left(\frac{V_{CC}}{2} + I_{IN} \times \frac{V_{CC}}{5} \times 100 \text{ mV/A} \right) \text{ V}$$

- **Power Supply:** The op-amp operates on a **3.3 V auxiliary supply**, ensuring compatibility with the MCU.
- **Offset Value:** At zero current, the output is **1.65 V**.

Key Advantages:

- Electrical Isolation
- Noise Reduction
- Wide Measurement Range
- Fast Response



LTspice Simulation Results

Input Current

The **input current** is a sine wave with **10 A peak** and **20 ms period**, representing typical converter operation.

Output Voltage

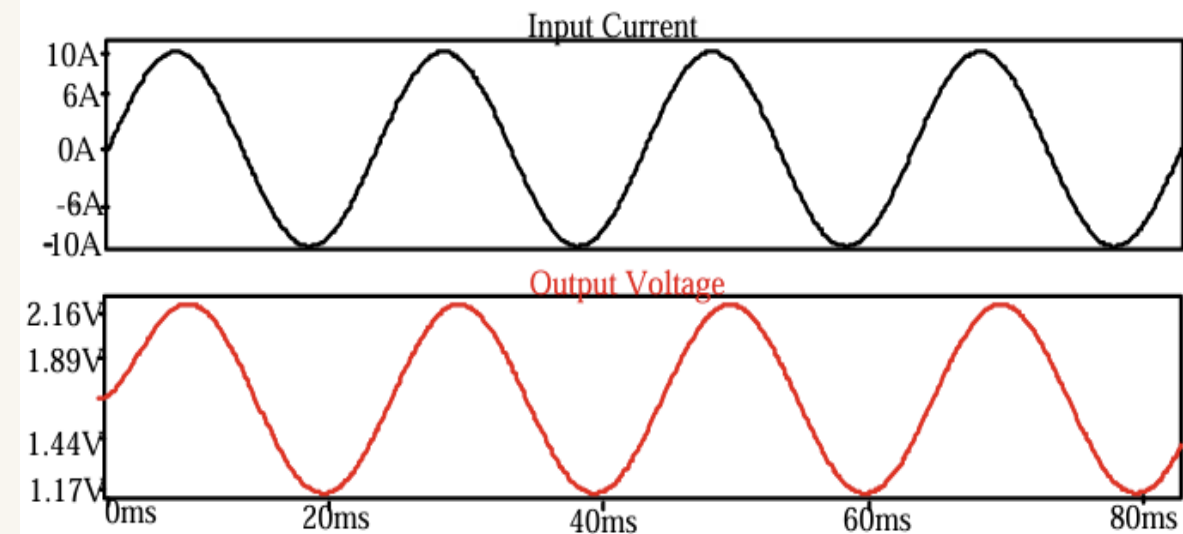
The **output voltage** is also sinusoidal (**~2.16 V peak**) and remains **in phase with the input current**.

The output stays within the **0–3.3 V range**, making it compatible with the microcontroller.

Verification

The results confirm correct **signal conditioning and gain settings**.

Overall, the simulation ensures **safe operation without exceeding MCU limits**



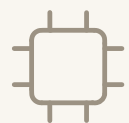
Complete System Schematic

- The system consists of **three identical input stages**, each using an **ACS730 current sensor** for three phase current measurement.
- Each stage includes components like a **20 A current source, shunt resistor, and capacitor** for signal conditioning.
- The sensor outputs are connected to **AD8608 op-amps** configured as **non-inverting amplifiers** with gain set by resistors.
- The outputs are further processed using an **AD8082 differential amplifier** for accurate signal conversion.
- A **low-pass RC filter** is applied at the final stage to remove noise before sending signals to the MCU.
- The system enables **independent sensing of each phase**, ensuring balanced monitoring.
- **Signal conditioning circuits** provide proper amplification and filtering for accurate measurement.
- All components, including **fault detection circuits**, are integrated on the **main PCB** for reliable operation.

Current Sensor System Architecture



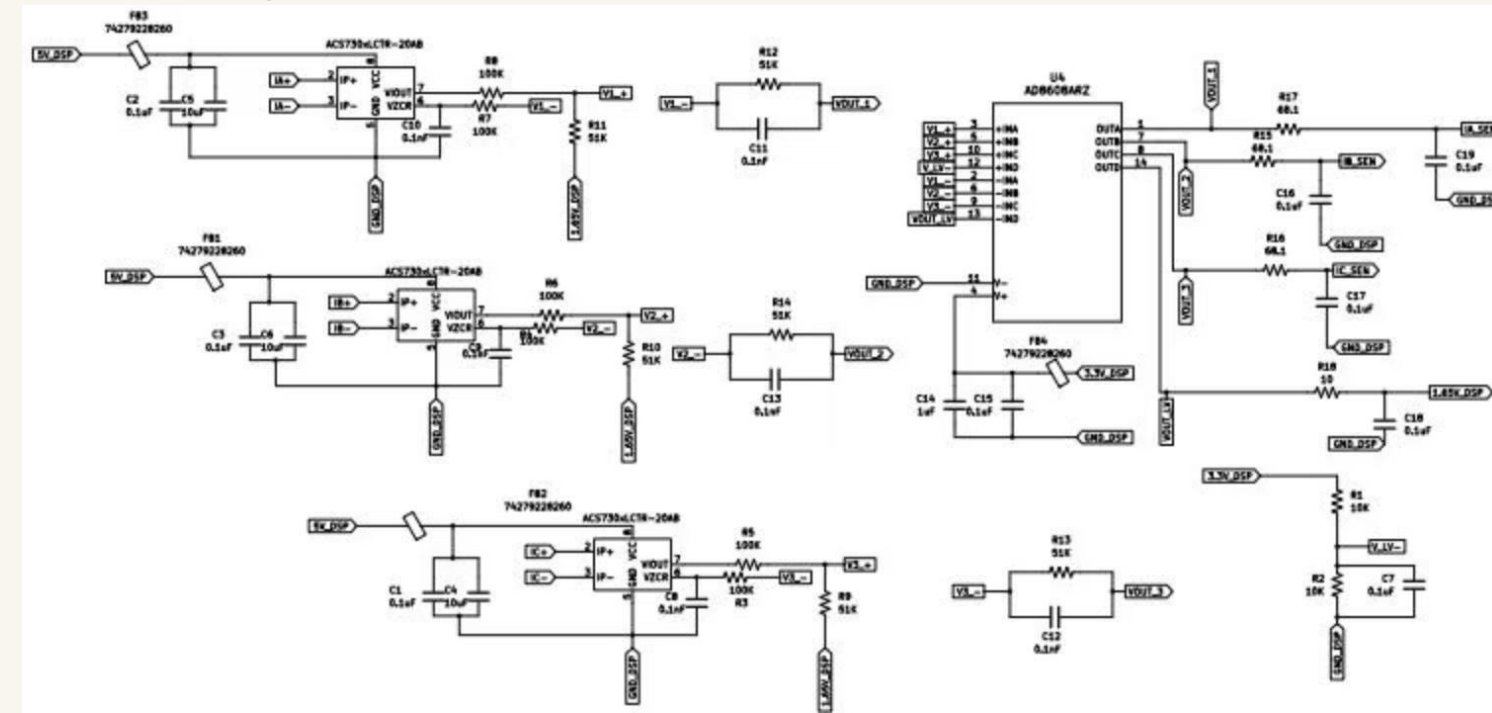
Three-Phase Input



Signal Conditioning



PCB Integration



THANK YOU